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Functional principal component models for sparse and irregularly spaced data by Bayesian inference

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ABSTRACT

The area of functional principal component analysis (FPCA) has seen relatively few contributions from the Bayesian inference. A Bayesian method in FPCA is developed under the cases of continuous and binary observations for sparse and irregularly spaced data. In the proposed Markov chain Monte Carlo (MCMC) method, Gibbs sampler approach is adopted to update the different variables based on their conditional posterior distributions. In FPCA, a set of eigenfunctions is suggested under Stiefel manifold, and samples are drawn from a Langevin–Bingham matrix variate distribution. Penalized splines are used to model mean trajectory and eigenfunction trajectories in generalized functional mixed models; and the proposed model is casted into a mixed-effects model framework for Bayesian inference. To determine the number of principal components, reversible jump Markov chain Monte Carlo (RJ-MCMC) algorithm is implemented. Four different simulation settings are conducted to demonstrate competitive performance against non-Bayesian approaches in FPCA. Finally, the proposed method is illustrated to the analysis of body mass index (BMI) data by gender and ethnicity.

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1. Introduction

To date there is a vast literature on principal component analysis (PCA) and functional data analysis (FDA). Functional principal component analysis (FPCA) is a technique that defines a set of smooth trajectories as an expansion of orthonormal eigenfunctions and functional principal component scores [36]. The Bayesian approach provides a principal alternative to asymptotic inference. However, Bayesian methods to FDA especially FPCA have been relatively absent from the literature [48,51]. Bayesian FPCA is still in its infancy and further development is needed. An important objective of this article is to propose a new Bayesian approach that could be added to the existing literature, which will enrich the different Bayesian applications in the area of FPCA.

Bayesian inference is a process of producing statistical inference taking a Bayesian point of view. Some computational difficulties can arise from Bayesian inference problem, which could be solved by either Markov chain Monte Carlo (MCMC) or Variational Inference (VI) methods [7]. The Gibbs sampler method in MCMC is based on the assumption that,

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even if the joint probability is intractable, the conditional distribution of a single dimension given the others can be computed [16,21]. When conditional distributions involved in Gibbs sampler is far too complex to be obtained, Metropolis-Hasting method can then be used [16,21]. The VI method consists in finding the best approximation of a distribution among a parametrized family. More specifically, the idea is to define a parametrized family of distributions and to optimize over the parameters. According to the review by Blei *et al.* [7], in general, MCMC is suited to smaller datasets and scenarios where researchers happily pay a heavier computational cost for more precise samples; VI is suited to large datasets and scenarios where researchers want to quickly explore many models. However, the relative accuracy of MCMC and VI is still unknown, which needs further exploration.

Some researchers [8,11,12,33] have applied Bayesian inference via MCMC methods in PCA. However, their estimations of eigenvectors were not incorporated into the Bayesian inference. These methods simply fix the eigenvectors, then perform MCMC without iterations of optimizing eigenvectors. Hence, Bayesian analysis for eigenvectors is reduced to regular PCA or maximum likelihood PCA in the procedure of data analysis. Some of them [14,20] incorporated the estimator of constrained eigenvectors into Bayesian inference, which is under Stielfel manifold [10].

Some other researchers considered that Bayesian inference for eigenvectors is based on VI methods to derive an approximate posterior distribution. In order to reduce the computing cost, some of them [6,13] elaborated VI in Bayesian PCA without explicitly requiring orthonormality constraint in the eigenvectors. If the model does not impose orthogonality restrictions, then the model inference is not identical to PCA. In classical PCA, orthonormal eigenvector span the best approximating subspaces. Hence, orthogonality constraints were imposed in the VI models [30,44].

FPCA is currently under intense methodological research and is a particular case of FDA. FPCA is PCA applied to functions instead of vectors, but functions differ from vectors by continuity and/or smoothness. It is particularly useful to reduce data dimension and accelerate the associated algorithms [11,12]. There is a large volume of recent work on FPCA. See [19,29,55] for kernel-based FPCA approaches; and [22,56,60,61] for spline-based FPCA methods. All these articles are concerned with the Gaussian type of functional data and cannot be used for generalized longitudinal trajectories. Hall *et al.* [19] proposed to model non-Gaussian longitudinal data by generalized linear mixed models, where the FPCA can be performed with respect to some latent random process.

These are the only references on Bayesian methods for principal components of functional data. Van Der Linde [51,52] proposed Bayesian inference via VI for generalized FPCA. It demonstrated that an adapted version of VI can be applied to both Gaussian process and latent random process. These VI methods are comparable with non-Bayesian approaches [19,55]. Suarez and Ghosal [48] and Jauch *et al.* [24] proposed MCMC methods in FPCA for dense and evenly spaced data.

In longitudinal data analysis, the individuals were usually measured in a fine grid of equally spaced time locations. However, in some cases, the measurements of the individuals did not follow the exactly scheduled time locations and might miss some locations. Hence, the actual time locations for the measurements of any one individual can be sparse and irregularly spaced and vary from one to another. When the measurements are under spare and irregularly spaced time locations, relatively less information is often available, which brings inherent difficulties in data analysis.

In Bayesian FDA, the methodology is applicable for densely or sparsely observed functional data, with automatic prediction or imputation at unobserved points [27,42]. However, most of the Bayesian FPCA methods typically focus on densely observed functional data. For example, the MCMC methods proposed by Jauch *et al.* [24], Suarez and Ghosal [48] are mainly for the analysis of the dense and evenly spaced data. Although their methods 'can easily accommodate certain types of missing data and can be flexibly modified or extended', there are no further details and discussions about how to handle sparse and irregularly spaced data with missing values. To the best of the author's knowledge, the proposed Bayesian method in this article is the first attempt to use fully Bayesian approach in FPCA on spare and irregularly spaced data.

The proposed Bayesian method is based on the reduced rank mixed-effects model in FPCA [22]. The reduced rank fitting procedure considers a rank constraint on the principal component scores, which interprets FPCA as a mixed-effects model for sparse and irregularly spaced data. The estimation procedure relies on a basis representation of the eigenfunctions and the principal component scores, and imposes the smoothness by a fixed number of basis functions [56,61]. The original B-spline basis functions are not orthonormal, therefore, the procedure prescribed by Ye *et al.* [56] is employed to orthogonalize them. It suggests that the performance of the penalized spline estimators is mainly controlled by the penalty parameters and is not sensitive to the choice of the number of the spline basis functions [13,56,61]. It is preferable to control smoothness by choosing a moderate number of basis functions for the analysis of the sparse and irregularly spaced data [61]. Moreover, in the reduced rank FDA for sparse data [22,56,61], a low-rank covariance function is assumed, meaning that only a small number of functional principal components are needed to represent it via the Karhunen-Loève expansion. Here Karhunen-Loève expansion is a representation of a stochastic process as an infinite linear combination of orthogonal functions. It is an eigenvector transformation for the trajectories, and is closely related to PCA.

In this article, a few new MCMC algorithms are developed in FPCA under the cases of continuous and binary observations for sparse and irregularly spaced data. Gibbs sampler is an alternating conditional sampling method in MCMC. It constructs a dependent sequence of parameter values whose distribution converges to the target joint posterior distribution [15,21]. In the proposed MCMC method, Gibbs sampler approach is adopted to update the different variables based on their conditional posterior distributions. A set of eigenfunctions is suggested under Stiefel manifold, and samples are drawn from a Langevin-Bingham matrix variate distribution. Penalized splines are used to model mean trajectory and eigenfunction trajectories in generalized functional mixed models; and the proposed model is casted into a mixed-effects model framework for Bayesian inference.

In the proposed generalized functional mixed model, binary response data are considered. Common methods for modeling binary outcomes include logistic and probit models, and they generally give similar results. Logistic model uses the logistic probability distribution function whereas probit model uses the truncated normal probability distribution function [39]. Although logistic model is used for binary data in many applications, the probit model is adopted in the method as it is much easier to implement with MCMC [9,39]. The probit regression model for binary responses has an underlying normal regression structure on latent continuous data. Given posterior distribution of the parameters, latent continuous data can be simulated from a suitable truncated normal distribution.

The choice of the number of PC's is an important sub-problem in FPCA. RJ-MCMC is an extension of Metropolis Hastings algorithm, and can perform parameter estimation and model selection simultaneously within the same framework. RJ-MCMC approach avoids computing marginal data densities by treating the number of the components as a parameter. RJ-MCMC becomes standard in Bayesian work with computing models with different number of parameters. Some researchers have applied RJ-MCMC for model selection in finite Gaussian mixture model, regression, factor analysis and PCA [26,31,37,47,59]. RJ-MCMC is used to select the number of principal components in FPCA. The method proposed by Suarez and Ghosal [48] is a VI-based model relying on covariance function expansion, where each pair of the number of basis function and the number of PCs needs be estimated in RJ-MCMC, causing the increase of computational burden when analyzing the dense data. Although the proposed method uses RJ-MCMC too, the computing cost is greatly reduced when the number of basis functions is fixed and the number of principal components is small for sparse data.

The article is organized as follows: Section 2 provides general descriptions of the method. Section 3 describes simulated and real data analysis to test the proposed method. Section 4 draws final conclusions. Appendix A provides technical details of the method, and Appendix B provides details of the simulation studies.

2. Methods

2.1. Formulas of FPCA

2.1.1. Generalized functional mixed models

In longitudinal data analysis, functional mixed-effects model is a traditional way for model fitting, prediction and inference when data is considered as functional. FPCA is one of the functional mixed-effects models archives two goals simultaneously: smoothing and dimension reduction. The objectives of FPCA are to revealing the overall trajectories from the measurements; uncover individual-specific variations and extract their dominant patterns.

Assume that longitudinal observations are realizations of canonical exponential family [32] and consider two different types of trajectories: one is Gaussian trajectories $\{Y_i(t_{ij}),\ t_{ij} \in \mathcal{T},\ j = 1, \ldots, n_i\}$, where $\mathcal{T}$ is a time interval for each individual i ($i = 1, \ldots, \mathcal{N}$); the other is binary trajectories $\{S_i(t_{ij}),\ t_{ij} \in \mathcal{T},\ j = 1, \ldots, n_i\}$ with $S_i(t_{ij}) = 1$ if the ith subject is active on time t_{ij} and $S_i(t_{ij}) = 0$ otherwise.

It is further assumed that $Y_i(t)$ is related to latent longitudinal process $X_i(t)$ with a standard Karhunen-Loève expansion

$$X_i(t) = \mu(t) + \psi(t)\xi_i, \quad \text{for } t \in \mathcal{T}, \tag{1}$$

where $\mu(t) = E\{X_i(t)\}$ is the mean function; $\psi = (\psi_1, \ldots, \psi_p)$ is a vector of orthonormal functions also known as the eigenfunctions in functional PCA; $\xi_i = (\xi_{i1}, \ldots, \xi_{ip})^T \sim \text{Normal}(\mathbf{0}, D_\xi)$ are the principal component scores, $D_\xi = \text{diag}(d_1, \ldots, d_p)$ and $d_1 \geq \ldots \geq d_p > 0$ are the eigenvalues. Given $X_i(t)$, Gaussian trajectories $Y_i(t)$ is independent of any $X_i(t_{ij}), j = 1, \ldots, n_i$. Let $f(Y_i|X_i)$ be the conditional distribution of $Y_i(t)|X_i(t)$, then $f(Y_i|X_i) \sim \text{Normal}(\mathbf{0}, \sigma_\varepsilon^2 I_{n_i})$, where I_{n_i} is a $n_i \times n_i$ identity matrix.

A liability augmentation process is considered for binary trajectories via the probit function. Binary outcomes $S_i(t_{ij})$, $i = 1, \ldots, \mathcal{N}$ and $j = 1, \ldots, n_i$, have an underlying truncated normal structure on latent continuous data. The liability $Y_i(t_{ij})$ is treated as an usual continuous character and is described to be a continuous model with normal distribution. Hence, $f(Y_i|X_i) \sim \text{Normal}(\mathbf{0}, \sigma_\varepsilon^2 I_{n_i})$. Since latent liability Y_i is unobserved, residual variance σ_ε^2 can be set arbitrarily [45]. It usually sets $\sigma_\varepsilon^2 = 1$ for simplification [39].

The distribution for binary outcomes S_i given the liability Y_i is

$$f\{S_i(t_{ij})|Y_i(t_{ij})\} = \prod_{j=1}^{n_i} \left[\mathbb{1}\{Y_i(t_{ij}) > 0\}\mathbb{1}\{S_i(t_{ij}) = 1\} + \{\mathbb{1}Y_i(t_{ij}) < 0\}\mathbb{1}\{S_i(t_{ij}) = 0\}\right], \quad (2)$$

where $\mathbb{1}(\cdot)$ is an indicator function [1,45].

2.1.2. Reduced rank model based on penalized B-splines

In some studies, trajectories were measured at spare and irregularly spaced time locations, which can differ widely across individuals. Hence, James *et al.* [22] proposed new techniques for handling this case by a reduced rank model. This model can be interpreted as a special case of mixed-effects model, where there is a rank constraint on the covariance matrix. The reduced rank model involves estimating fewer parameters in FPCA and as a result the fitted trajectories are less variable and generally more accurate. It enables predictions of individual trajectories even if few measurements are available in each individual. Hence, reduced rank model is more suitable for sparse data analysis.

A flexible approach in reduced rank model is to use a moderate number of B-spline basis functions and apply a roughness penalty to regularize the fitted trajectories [61]. In FPCA, as long as the moderate number of B-spline basis functions is included, the placement of knots is not critical. The B-spline basis functions are flexible enough to capture the pattern of the data. To simplify the data analysis, in the proposed method, the B-spline basis on equally spaced knots is considered.

The unknown $\mu(t)$ and $\psi(t)$'s are approximated by B-splines. Let $\mathscr{B}(t) = \{\mathscr{B}_1(t), \ldots, \mathscr{B}_q(t)\}^T$ be a q-dimensional B-spline basis defined by equally spaced knots in $\mathcal{T}$, θ_μ be a $q \times 1$ vector and $\Theta_\psi = (\theta_{\psi 1}, \ldots, \theta_{\psi p})$ be a $q \times p$ matrix of spline coefficients, then unknown functions are represented as $\mu(t) = \mathscr{B}(t)^{\mathrm{T}}\theta_\mu$ and $\psi^{\mathrm{T}}(t) = \mathscr{B}(t)^{\mathrm{T}}\Theta_\psi$. General recommendation for choosing q in penalized spline literature is to choose a relatively large number $q \gg p$, and have smoothness of estimated functions regularized by the roughness penalties [39]. The original B-spline basis functions are not orthonormal. As the data-driven feature of the functional data is the projections of the sparse data onto orthonnormal basis functions [29], the procedure prescribed by Ye *et al.* [56] was employed to orthogonalize the basis functions. Hence, $\int \mathscr{B}(t)\mathscr{B}(t)^T \, \mathrm{d}t = I_q$, where I_q is a $q \times q$ identity matrix. Under this construction, orthonormal constraints on $\psi(t)$ translate to constraints on the coefficients, i.e. $\Theta_\psi^T \Theta_\psi = I_p$. Then the reduced rank model for the latent process takes a form as

$$X_i(t) = \mathscr{B}(t)^T\theta_\mu + \mathscr{B}(t)^T\Theta_\psi\xi_i, \quad \text{subject to } \Theta_\psi^T\Theta_\psi = I_p. \quad (3)$$

For Gaussian trajectories Y_i,

$$Y_i = B_i\theta_\mu + B_i\Theta_\psi\xi_i + \varepsilon_i, \quad (4)$$

where $B_i = \{\mathscr{B}(t_{i1}), \ldots, \mathscr{B}(t_{in_i})\}^T$ is the design matrix by interpolating the basis functions on the observation time and $\varepsilon_i \sim \text{Normal}(\mathbf{0}, \sigma_\varepsilon^2 I_{n_i})$.

If considering an additional fixed covariate $\mathscr{Z}(t)$ in the model, and $X_i(t) = \mathscr{Z}(t)^T\theta_\eta + \mathscr{B}(t)^T\theta_\mu + \mathscr{B}(t)^T\Theta_\psi\xi_i$, then

$$Y_i = Z_i\theta_\eta + B_i\theta_\mu + B_i\Theta_\psi\xi_i + \varepsilon_i, \tag{5}$$

where $Z_i = \{\mathscr{Z}(t_{i1}), \ldots, \mathscr{Z}(t_{in_i})\}^T$. For binary trajectories S_i, unknown liability Y_i has the same Equation (5) with $\varepsilon_i \sim \text{Normal}(\mathbf{0}, I_{n_i})$. Let $\mathbf{Y} = (Y_1, \ldots, Y_\mathcal{N})$, the conditional likelihood function of $\mathbf{Y}$ is

$$\begin{aligned} f(\mathbf{Y}|\xi_i, \sigma_\varepsilon^2, \theta_\mu, \theta_\eta, \Theta_\psi, h_\mu, h_\psi) &= \left\{\prod_{i=1}^{\mathcal{N}} f(Y_i)\right\} \\ &\propto \left(\frac{1}{\sigma_\varepsilon^2}\right)^{\sum_{i=1}^{\mathcal{N}} \frac{n_i}{2}} \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}} \| Y_i - Z_i\theta_\eta - B_i\theta_\mu - B_i\Theta_\psi\xi_i \|^2\right\}\right]. \end{aligned} \tag{6}$$

With functional data analysis, observations of functions at different possibly irregularly spaced points can be handled. A roughness penalty is usually introduced in the functional data analysis, which is equivalent to a smoothness prior and is an obvious option in Bayesian analysis. To regularize the nonparametric estimators, the roughness penalties are imposed on the L^2 norm of the second derivatives [39]. Define $\mathcal{J}_\mathcal{B} = \int \mathscr{B}''(t)\mathscr{B}''(t)^T \, dt$, then

$$\int \left\{\mu''(t)\right\}^2 \, dt = \theta_\mu^T \mathcal{J}_\mathscr{B}\theta_\mu, \quad \int \left\{\psi_k''(t)\right\}^2 \, dt = \theta_{\psi l}^T \mathcal{J}_\mathscr{B}\theta_{\psi l},$$

where $\theta_{\psi l}$ is lth column of Θ_ψ and $l = 1, \ldots, p$. The penalized likelihood function of $\mathbf{Y}$ is

$$\begin{aligned} &f(\mathbf{Y}|\xi_i, \sigma_\varepsilon^2, \theta_\mu, \theta_\eta, \Theta_\psi, h_\mu, h_\psi) \\ &= \left\{\prod_{i=1}^{\mathcal{N}} f(Y_i)\right\} \exp\left\{-\frac{1}{2\sigma_\varepsilon^2}\left(\frac{1}{2}h_\mu\theta_\mu^T\mathcal{J}_\mathcal{B}\theta_\mu + \frac{1}{2}h_\psi\sum_{l=1}^{p}\theta_{\psi l}^T\mathcal{J}_\mathcal{B}\theta_{\psi l}\right)\right\} \\ &\propto \left(\frac{1}{\sigma_\varepsilon^2}\right)^{\sum_{i=1}^{\mathcal{N}} \frac{n_i}{2}} \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}} \| Y_i - Z_i\theta_\eta - B_i\theta_\mu - B_i\Theta_\psi\xi_i \|^2\right\}\right] \\ &\exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\frac{1}{2}h_\mu\theta_\mu^T\mathcal{J}_\mathcal{B}\theta_\mu + \frac{1}{2}h_\psi\sum_{l=1}^{p}\theta_{\psi l}^T\mathcal{J}_\mathcal{B}\theta_{\psi l}\right\}\right], \end{aligned} \tag{7}$$

where h_μ, h_ψ are regularization tuning parameters.

Let $A_i = (Z_i, B_i)$ and $\theta = \binom{\theta_\eta}{\theta_\mu}$, then

$$A_i\theta = Z_i\theta_\eta + B_i\theta_\mu. \tag{8}$$

Define $\mathcal{J}_\mathcal{A} = \text{diag}(\mathbf{0}_{g\times g}, \mathcal{J}_\mathcal{B})$, where $\mathbf{0}_{g\times g}$ is a $g \times g$ matrix with all entries of zero and g to be the number of the covariates, then $\theta^T\mathcal{J}_\mathcal{A}\theta = \theta_\mu^T\mathcal{J}_\mathcal{B}\theta_\mu$. The penalized likelihood function of $\mathbf{Y}$ becomes

$$\begin{aligned} & f(\mathbf{Y}|\xi_i, \sigma_\varepsilon^2, \theta, \Theta_\psi, h_\mu, h_\psi) \\ & \quad \propto \left(\frac{1}{\sigma_\varepsilon^2}\right)^{\sum_{i=1}^{\mathcal{N}} \frac{n_i}{2}} \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}} \parallel Y_i - A_i\theta - B_i\Theta_\psi\xi_i \parallel^2\right\}\right] \\ & \quad \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\frac{1}{2}h_\mu\theta^T\mathcal{J}_\mathcal{A}\theta + \frac{1}{2}h_\psi\sum_{l=1}^{p}\theta_{\psi l}^T\mathcal{J}_\mathcal{B}\theta_{\psi l}\right\}\right]. \end{aligned} \tag{9}$$

2.2. Bayesian description of singular value decomposition

Singular value decomposition (SVD) is simply matrix representation of a transformation of principal component analysis (PCA) [25]. Karhunen-Loève expansion is an eigenvector transformation for the trajectories, which is PCA in a continuous domain. When the time points are considered as discrete, there is a relationship between Karhunen-Loève expansion and SVD.

Let $\xi_i = E_\xi\Theta_{\xi i}^T$, the matrix $\Theta_\psi\xi$ is further decomposed as $\Theta_\psi E_\xi\Theta_\xi^T$ based on the idea of SVD, where both Θ_ψ and Θ_ξ are $q \times p$ and $\mathcal{N} \times p$ matrices respectively, each has orthogonal columns, i.e. $\Theta_\psi^T\Theta_\psi = I_p$, $\Theta_\xi^T\Theta_\xi = \mathcal{N}I_p$; and E_ξ is a diagonal matrix with rank p such that $e_1 \geq \ldots \geq e_p > 0$ and $e_l = \sqrt{d_l}$ for $l = 1, \ldots, p$. After assigning uniform prior distributions to Θ_ψ and Θ_ξ on space of Stiefel manifold, Bayesian inference on Θ_ψ and Θ_ξ can be conducted under conditional posterior distributions [14,20,44]. The penalized likelihood function of $\mathbf{Y}$ is

$$\begin{aligned} & f(\mathbf{Y}|\Theta_\psi, \Theta_\xi, E_\xi, \sigma_\varepsilon^2, \theta, h_\mu, h_\psi) \\ & \quad = \left\{\prod_{i=1}^{\mathcal{N}} f(Y_i)\right\} \exp\left\{-\frac{1}{2\sigma_\varepsilon^2}\left(\frac{1}{2}h_\mu\theta^T\mathcal{J}_\mathcal{A}\theta + \frac{1}{2}h_\psi\sum_{l=1}^{p}\theta_{\psi l}^T\mathcal{J}_\mathcal{B}\theta_{\psi l}\right)\right\} \\ & \quad \propto \left(\frac{1}{\sigma_\varepsilon^2}\right)^{\sum_{i=1}^{\mathcal{N}} \frac{n_i}{2}} \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}} \parallel Y_i - A_i\theta - B_i\Theta_\psi E_\xi\Theta_{\xi i}^T \parallel^2\right\}\right] \\ & \quad \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\frac{1}{2}h_\mu\theta^T\mathcal{J}_\mathcal{A}\theta + \frac{1}{2}h_\psi\sum_{l=1}^{p}\theta_{\psi l}^T\mathcal{J}_\mathcal{B}\theta_{\psi l}\right\}\right], \end{aligned} \tag{10}$$

where h_μ, h_ψ are regularization tuning parameters.

2.3. Bayesian inference in the fixed rank model

A fixed rank model is considered at first, where number of principal components p is predetermined.

2.3.1. Gibbs sampler for continuous outcomes

In a Gibbs sampler framework, all model parameters are considered to be drawn from appropriate prior distributions. Assume the prior distributions for different unknown variables are independent of a priori, then prior distributions for $\Theta_\psi, \Theta_\xi, E_\xi, \sigma_\varepsilon^2, \theta, h_\mu$ and h_ψ could be calculated, respectively.

2.3.1.1. Prior for $\boldsymbol{\Theta_\psi}$. Note that Θ_ψ is a $q \times p$ matrix with orthogonal columns whose rank is p, i.e. $\Theta_\psi^T \Theta_\psi = I_p$. A set of such matrices, denoted by $V_{p,q}$, is Stiefel manifold [10,18]

$$V_{p,q} = \{\Theta_\psi(q \times p) : \Theta_\psi^T \Theta_\psi = I_p\}. \tag{11}$$

To reflect absence of any additional prior knowledge regarding Θ_ψ, a uniform distribution is chosen as the prior distribution [14,20]

$$f(\Theta_\psi) = \frac{1}{\mathrm{vol}(V_{p,q})} \mathbb{1}(\Theta_\psi \in V_{p,q}), \tag{12}$$

where $\mathbb{1}(\cdot)$ is an indicator function, and $\mathrm{vol}(V_{p,q}) = \int_{V_{p,q}} \Theta_\psi^T d\Theta_\psi$ is surface area (referred to as the volume) of the Stiefel manifold $V_{p,q}$. The columns of Θ_ψ (i.e. $\Theta_{\psi,[\cdot,l]}, l = 1, \ldots, p$) will be iteratively sampled conditionally upon others (i.e. $\Theta_{\psi[\cdot,-l]}$). Thus

$$f(\Theta_{\psi,[\cdot,l]} | \Theta_{\psi[\cdot,-l]}) \propto \mathbb{1}(\Theta_{\psi,[\cdot,l]} \in V_{1,q} | \Theta_{\psi[\cdot,-l]}). \tag{13}$$

$\Theta_{\psi,[\cdot,l]}$ is further constructed as $\Theta_{\psi,[\cdot,l]} = M_l \nu_l$ such that

$$f(\nu_l | \Theta_{\psi[\cdot,-l]}) \propto \mathbb{1}(\nu_l \in V_{1,(q-p+1)} | \Theta_{\psi[\cdot,-l]}), \tag{14}$$

where ν_l is uniform from the unit sphere R^{q-p+1} (i.e. $V_{1,(q-p+1)}$) and M_l is a $q \times (q - p + 1)$ matrix whose columns form an orthonormal basis for null space of $\Theta_{\psi[\cdot,-l]}$, i.e.

$$M_l^T M_l = I_{(q-p+1)}, \quad M_l^T \Theta_{\psi[\cdot,-l)} = \mathbf{0}_{(q-p+1)\times(p-1)}. \tag{15}$$

2.3.1.2. Prior for $\boldsymbol{\Theta_\xi}$. Note that Θ_ξ is a $\mathcal{N} \times p$ matrix with orthogonal columns whose rank is p. A set of such matrices, denoted by $V_{p,\mathcal{N}}$, is Stiefel manifold

$$V_{p,\mathcal{N}} = \{\Theta_\xi(\mathcal{N} \times p) : \Theta_\xi^T \Theta_\xi = I_p\}. \tag{16}$$

A uniform distribution is chosen as the prior distribution,

$$f(\Theta_\xi) = \frac{1}{\mathrm{vol}(V_{p,\mathcal{N}})} \mathbb{1}(\Theta_\xi \in V_{p,\mathcal{N}}), \tag{17}$$

where $\mathbb{1}(\cdot)$ is an indicator function, and $\mathrm{vol}(V_{p,\mathcal{N}}) = \int_{V_{p,\mathcal{N}}} \Theta_\xi^T d\Theta_\xi$ is surface area (volume) of Stiefel manifold $V_{p,\mathcal{N}}$. Columns of Θ_ξ (i.e. $\Theta_{\xi,[\cdot,l]}, l = 1, \ldots, p$) will be iteratively sampled conditionally upon others (i.e. $\Theta_{\xi[\cdot,-l]}$)

$$f(\Theta_{\xi,[\cdot,l]}|\Theta_{\xi[\cdot,-l]}) \propto \mathbb{1}(\Theta_{\xi,[\cdot,l]} \in V_{1,q}|\Theta_{\xi[\cdot,-l]}). \tag{18}$$

$\Theta_{\xi,[\cdot,l]}$ is further constructed as $\Theta_{\xi,[\cdot,l]} = M_{\xi l}\nu_{\xi l}$ such that

$$f(\nu_{\xi l}|\Theta_{\xi[\cdot,-l]}) \propto \mathbb{1}(\nu_{\xi l} \in V_{1,(\mathcal{N}-1)}|\Theta_{\xi[\cdot,-l]}), \tag{19}$$

where $\nu_{\xi l}$ is uniform from unit sphere $R^{\mathcal{N}-p+1}$ (i.e. $V_{1,(\mathcal{N}-p+1)}$) and $M_{\xi l}$ is a $\mathcal{N} \times (\mathcal{N} - p + 1)$ matrix whose columns form an orthonormal basis for null space of $\Theta_{\xi[\cdot,-l]}$, i.e.

$$M_{\xi l}^T M_{\xi l} = I_{(\mathcal{N}-p+1)}, \quad M_{\xi l}^T \Theta_{\xi[\cdot,-l)} = \mathbf{0}_{(\mathcal{N}-p+1)\times(p-1)}. \tag{20}$$

2.3.1.3. *Prior for E_ξ.*

Considering $e_l = E_{\xi[l,l]}$ with $l = 1, \ldots, p$, prior of e_l is

$$f(e_l) \propto \text{Normal}\left\{-\frac{1}{2\sigma_e^2}(e_l - \mu_e)^2\right\}, \tag{21}$$

where μ_e and σ_e^2 are hyper-parameters [16,20].

2.3.1.4. *Prior for μ_e and σ_e^2.*

Considering μ_e is normally distributed, i.e.

$$f(\mu_e) \propto \exp\left\{-\frac{1}{2\nu_{eo}^2}(\mu_e - \mu_{eo})^2\right\}, \tag{22}$$

and

$$f(\frac{1}{\sigma_e^2}) \sim \text{Gamma}(c_e, d_e) = \left(\frac{1}{\sigma_e^2}\right)^{c_e-1} \exp\left(-d_e\frac{1}{\sigma_e^2}\right), \tag{23}$$

where $c_e = 1$ and d_e is a small number. The hyper-parameters μ_{eo}, ν_{eo}^2, c_e and d_e, are presented as fixed values in the analysis [16,20].

2.3.1.5. *Prior for σ_ε^2.*

Prior distribution for noise variance [14,28] is

$$f\left(\frac{1}{\sigma_\varepsilon^2}\right) \propto \text{Gamma}(c_\varepsilon, d_\varepsilon) = \left(\frac{1}{\sigma_\varepsilon^2}\right)^{c_\varepsilon-1} \exp\left(-d_\varepsilon\frac{1}{\sigma_\varepsilon^2}\right), \tag{24}$$

where hyper-parameters $c_\varepsilon = 1$ and d_ε is a small number (e.g. 0.1).

2.3.1.6. *Prior for θ.*

Prior for θ [2,49] is

$$f(\theta|h_\mu, \sigma_\varepsilon^2) \propto \left\{-\frac{1}{2\sigma_\varepsilon^2}\left(\frac{1}{2}h_\mu\theta^T\mathcal{J}_A\theta\right)\right\}. \tag{25}$$

2.3.1.7. *Prior for h_μ and h_ψ.*

Priors of h_μ and h_ψ [2] are

$$f(h_\mu) \sim \text{Gamma}(c_\mu, d_\mu) \propto (h_\mu)^{c_\mu-1}\exp(-d_\mu h_\mu), \tag{26}$$

$$f(h_\psi) \sim \text{Gamma}(c_\psi, d_\psi) \propto (h_\psi)^{c_\psi-1}\exp(-d_\psi h_\psi). \tag{27}$$

The priors for h_μ and h_ψ must not be diffuse in order to obtain proper posteriors θ_μ and Θ_ψ. Common choice for the hyper-parameters is that $c_\mu = c_\psi = 1$ and small values (e.g. 0.005 or 0.0005) for d_μ and d_ψ [16,20].

2.3.1.8. Posterior distribution. The unknown parameters of the model are $\Theta = \{\Theta_\psi, \Theta_\xi, E_\xi, \sigma_\varepsilon^2, \theta, h_\mu, h_\psi\}$, and the hyper-parameters are $\Phi = \{\mu_e, \sigma_e^2\}$. Based on the specified prior distributions, joint posterior distribution of $\{\Theta, \Phi\}$ can be expressed as the hierarchical structure

$$f(\Theta, \Phi|\mathbf{Y}) \propto f(\mathbf{Y}|\Theta)f(\Theta|\Phi)f(\Phi), \tag{28}$$

where $f(\mathbf{Y}|\Theta)$ is the likelihood function of continuous outcome $\mathbf{Y} = (Y_1, \ldots . Y_\mathcal{N})$, functions $f(\Theta|\Phi) = f(E_\xi|\mu_e, \sigma_e^2)f(\sigma_\varepsilon^2|c_\varepsilon, d_\varepsilon)f(h_\mu|c_\mu, d_\mu)f(h_\psi|c_\psi, d_\psi)f(\Theta_\phi)f(\Theta_\xi)f(\theta)$, and $f(\Phi) = f(\mu_e|\mu_{eo}, \nu_{eo}^2)f(\sigma_e^2|c_e, d_e)$.

In the Bayesian approach, posterior samples are generated from joint posterior distribution of all unknown variables, which can provide comprehensive posterior inferences [15,58]. For the joint posterior distribution, conditional posterior distribution of each unknown variable can be derived by treating other unknown variables as constant and selecting the functions of interest. Because of the orthogonality constrain, posterior distribution of the eigenfunctions is found to be of the Langevin-Bingham type under Stiefel manifold. This is a matrix of normal distribution with orthogonality constrain on the matrix argument. Expressions of the conditional posterior distributions are given in Appendix A.

In particular, closed forms of Bayesian estimator of some parameters are difficult to obtain. The MCMC methods are used to generate samples that are asymptotically distributed according to the target distribution [14]. Gibbs sampler approach is adopted to update the different variables based on their conditional posterior distributions. The approach proceeds by iteratively sampling each parameter from its full conditional distribution. Each random variable (model parameter) is iteratively drawn from its conditional posterior distribution given the most recent values of all the other parameters. The algorithm starts from the initial values and proceeds to iteratively modify each of the unknowns in turn. After a large number of burn-in iterations, the Markov chain is stable. The algorithm of Gibbs sampler for continuous outcomes is as below:

Algorithm 2.1: (1) For $l = 1, \ldots, p$, sample parameters $\theta_{\psi l}$ (A13), $\theta_{\xi l}$ (A18), and e_l (A21) iteratively.
(2) Sample parameters σ_ε^2 (A24), θ (A27), h_μ (A29), and h_ψ (A30).
(3) Sample hyper-parameters μ_e (A33) and σ_e^2 (A36).
(4) Return to Step 1 until a pre-specified number of iterations has been reached.

2.3.2. Gibbs sampling for binary outcomes

To facilitate development of the Gibbs sampler, unobserved liability $\mathbf{Y}$ is included as an unknown variable in the model, which is known as the data augmentation approach [1,37]. Then the joint posterior distribution of the variables is

$$f(\mathbf{S}, \Theta, \Phi|\mathbf{Y}) \propto f(\mathbf{S}|\mathbf{Y})f(\mathbf{Y}|\Theta)f(\Theta|\Phi)f(\Phi). \tag{29}$$

Compared with Algorithm 2.1, the algorithm for the binary outcomes adds an additional update step to sample the liability $\mathbf{Y}$, and cancels the update of σ_ε^2 which is assumed to be fixed, i.e. $\sigma_\varepsilon^2 = 1$. The algorithm of Gibbs sampler for binary outcomes is as below:

Algorithm 2.2: (1) Sample $\mathbf{Y} = (Y_1, \ldots, Y_{\mathcal{N}})$ by the probit model. Every Y_i is drawn from a truncated normal distribution under $S_i = 1$ by (A7) or $S_i = 0$ by (A8).
(2) For $l = 1, \ldots, p$, sample parameters $\theta_{\psi l}$ (A13), $\theta_{\xi l}$ (A18), and e_l (A21) iteratively.
(3) Sample parameters θ (A27), h_μ (A29), and h_ψ (A30).
(4) Sample hyper-parameters μ_e (A33) and σ_e^2 (A36).
(5) Return to Step 1 until a pre-specified number of iterations has been reached.

2.4. RJ-MCMC for random rank model

The proper prior distributions for all parameters are selected, including the number of the principal components by RJ-MCMC algorithm, which performs birth and death moves in every reversible jump step. The birth and death moves regard each principal component as a point in a one-dimensional space [59], i.e. adding one PC to the model is regarded as one birth move, and removing one PC from the model is regarded as one death move. The prior distribution for number of principal components p is usually assumed to be truncated poisson with a constant λ, representing the mean of the number of principal components. Maximum number of principal components is controlled as $p_{\max}$ such that $1 \leq \lambda \leq p_{\max}$. Let $f(\text{birth})$ and $f(\text{death}) = 1 - f(\text{birth})$ be the proposed probabilities of choosing the types of birth and death moves, they usually satisfy the conditions such that $f(\text{birth}) = 0.5$ if $1 < p < p_{\max}$, $f(\text{birth}) = 1$ if $p = p_{\max}$, and $f(\text{birth}) = 0$ if $p = 1$. Following [43,59], algorithm of RJ-MCMC is as below:

Algorithm 2.3: (1) Update all parameters and hyper-parameters with fixed number of components using either Algorithms 2.1 or 2.2.
(2) Conditionally based on the other current parameters, select the number of principal components by making a random choice between birth move and death move.
    (1) Draw a uniform random variable $u \sim U(0, 1)$.
    (2) If $u \leq f(\text{birth})$, then perform the birth move.
    (3) If $u > f(\text{birth})$, then perform the death move.
(3) Return to Step 1 until a pre-specified number of iterations has been reached.

See Appendix A for the technical details of birth move and death move.

3. Numerical studies

3.1. Simulation settings

To illustrate performance of the suggested algorithms in Bayesian data analysis, four different simulation settings were conducted.

In simulation setting one, there are 100 simulations conducted for Gaussian longitudinal data. In every simulation study, $\mathcal{N} = 100$ independent individuals are simulated, with $n_i = 15 - 30$ observations irregularly spaced on time interval $\mathcal{T} = [0, 30]$. In simulation setting two, there are 100 simulations conducted for binary longitudinal data. In every simulation study of the binary data, $\mathcal{N} = 25$ independent individuals are simulated, with $n_i = 1 - 10$

observations irregularly spaced on time interval $\mathcal{T} = [0, 10]$. In simulation setting three, there are 150 simulations in total conducted for the Gaussian longitudinal data. The mean function and three eigenfunctions are the same as those in simulation setting one. There are six different simulation studies under $n_i = 1-5, 3-10, 6-15, 9-20, 12-15$ and 15–30 with fixed $\sigma_\epsilon^2 = 0.7^2$. In simulation setting four, there are 150 simulations in total conducted for the Gaussian longitudinal data. The mean function and three eigenfunctions are the same as those in simulation setting one. The measurement error $\varepsilon(t)$ is a Gaussian white noise process with variance σ_ϵ^2. There are six different simulation studies under $\sigma_\epsilon^2 = 0.5^2, 0.7^2, 0.9^2, 1.1^2, 1.3^2$ and 1.5^2.

Details of the simulation studies are described in Appendix B.

3.1.1. Results in simulation settings one and two

First, the RJ-MCMC method was run until the convergence of the MCMC outputs for the Gaussian longitudinal data and the binary longitudinal data respectively. Of the 100 continuous simulations, 76% picked up $p = 3$; of the 100 binary simulations, 69% picked up $p = 3$. Hence the algorithm selected the number of the principal components as 3 in most of the simulations, matching the results of the initial simulations, that is, there are 3 principal components in the data sets. Second, as the MCMC method does not provide one specific value on the number of principal components that should be retained in the analysis [59], the number of principal components was fixed as 3 for all continuous and binary data sets so that the results from Bayesian inference can be easily used to compare with those from frequentist inference. The non-Bayesian approaches from [19,55] were used for comparison through the simulation studies. In the final results, both average mean square error (MSE) were reported for the true curves and average square error (ASE) for the three principal component scores. Please see the definitions of MSE and ASE in [55]. The smaller the values of MSE and ASE are, the better the model is.

Specifically, comparisons are between the results and those from FPCA through conditional expectation (PACE) method for continuous responses [55], and through PACE with Generalized (non-Gaussian) Repeated Measurements (GRM) for the binary responses [19]. Table 1 and Figure 1 report the results from continuous responses. Table 2 and Figure 2 report the results from binary responses. It is seen that both methods show similar estimation performance and all estimates behave reasonably well. The comparison between the two methods indicates that the Bayesian inference produces slightly better estimates, i.e. the MSEs of general function, its baseline function and three eigenfunctions are smaller; the ASEs of three principal component scores are smaller.

Table 1. Comparison of the Bayesian method and the PACE method for the continuous responses.

Bayes	MSE	MSE(μ)	MSE(θ_{ψ_1})	MSE(θ_{ψ_2})	MSE(θ_{ψ_3})	ASE(ξ_1)	ASE(ξ_2)	ASE(ξ_3)
Mean	1.7012	0.0169	0.0518	0.0547	0.0730	35.3785	10.7206	2.7758
Std	0.9564	0.0112	0.0629	0.0614	0.0496	31.6746	9.0751	1.2829
PACE	MSE	MSE(μ)	MSE(θ_{ψ_1})	MSE(θ_{ψ_2})	MSE(θ_{ψ_3})	ASE(ξ_1)	ASE(ξ_2)	ASE(ξ_3)
Mean	2.6303	0.0156	0.0540	0.0609	0.0854	50.3736	13.4654	7.2882
Std	1.7193	0.0094	0.0622	0.0581	0.0512	51.1733	15.2082	9.1334

Note: Results show the mean and standard deviation for the MSE and ASE of 100 independent simulations.

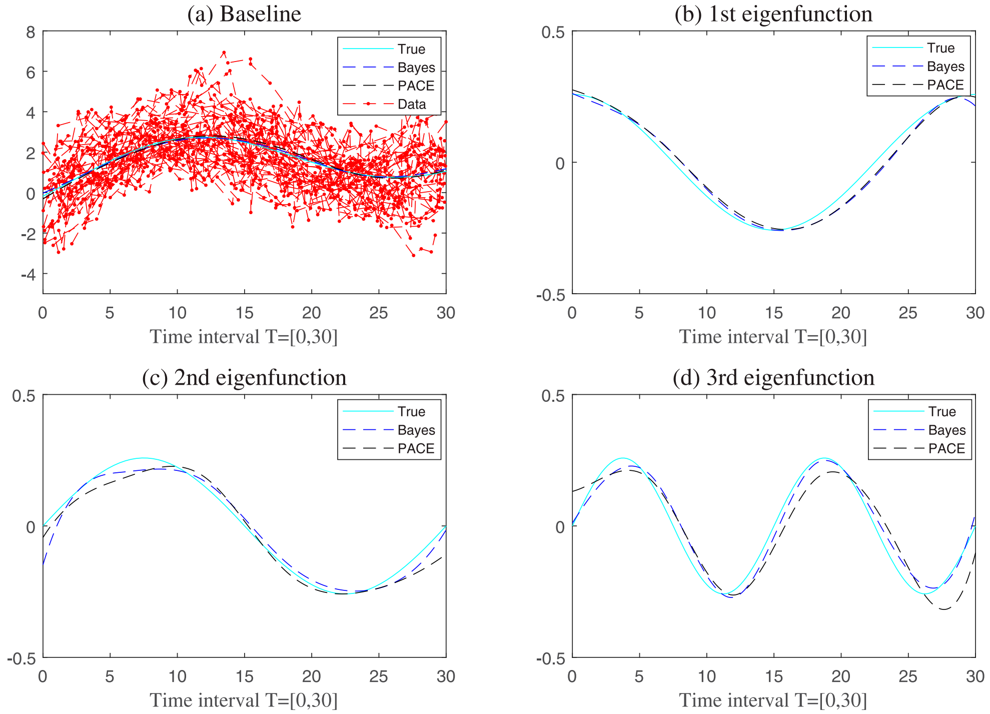

Figure 1. Three eigenfunctions for the continuous responses under one specific simulated data set on time interval $\mathcal{T} = [0, 30]$. Both Bayesian method and PACE method fit the shapes of the true curves very well. The data points in panel (a) are from Gaussian process.

Table 2. Comparison of the Bayesian method and the PACE-GRM method for the binary responses.

Bayes	MSE	MSE(μ)	MSE(θ_{ψ_1})	MSE(θ_{ψ_2})	MSE(θ_{ψ_3})	ASE(ξ_1)	ASE(ξ_2)	ASE(ξ_3)
Mean	4.8794	0.6681	0.1589	0.1551	0.1570	33.0765	11.5634	2.8386
Std	1.3271	0.5932	0.1320	0.1032	0.0714	16.6080	7.2348	1.6750
GRM	MSE	MSE(μ)	MSE(θ_{ψ_1})	MSE(θ_{ψ_2})	MSE(θ_{ψ_3})	ASE(ξ_1)	ASE(ξ_2)	ASE(ξ_3)
Mean	6.4241	0.6777	0.1807	0.1742	0.1752	34.1982	13.5418	4.8841
Std	2.4515	0.2614	0.1502	0.1024	0.0691	25.1855	8.3646	2.8032

Note: Results show the mean and standard deviation for the MSE and ASE of 100 independent simulations.

3.1.2. Results in simulation settings three and four

To make further comparisons between Bayesian method and PACE method, simulation settings three and four were conducted under different conditions. In order to make fair comparisons in different studies, the number of principal components was fixed as 3 when Bayesian method and PACE method were applied.

In simulation setting three, Panel (a) to Panel (f) in Figure 3 show the simulated sparse data points under $n_i = 1-5, 3-10, 6-15, 9-20, 12-15$ and 15–30 with fixed $\sigma_\epsilon^2 = 0.7^2$. It is seen that there are more densely points while n_i is increasing. The results for the comparison of Bayesian method and PACE method are shown in Figure 5.

In simulation setting four, Panel (a) to Panel (f) in Figure 4 show the simulated sparse data points under $\sigma_\epsilon^2 = 0.5^2, 0.7^2, 0.9^2, 1.1^2, 1.3^2$ and 1.5^2 with fixed $n_i = 9-20$. It is seen

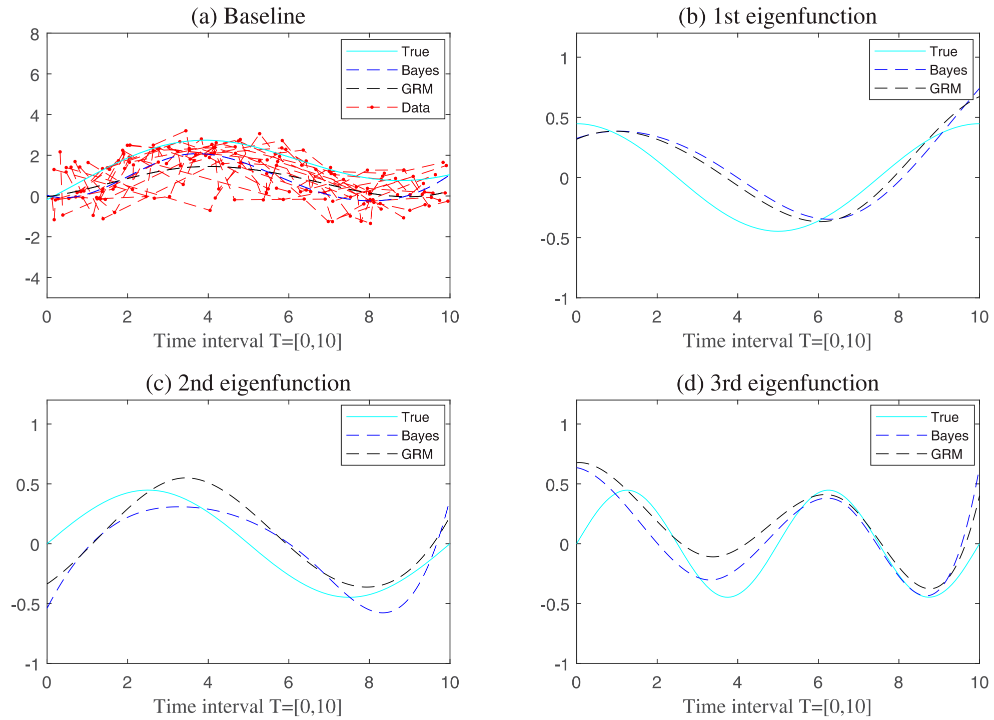

Figure 2. Three eigenfunctions for the binary responses under one specific simulated data set on time interval $\mathcal{T} = [0, 10]$. Both Bayesian method and PACE-GRM method fit the shapes of the true curves well. The data points in panel (a) are the latent values from the truncated Normal distribution.

that the data points are more spread while σ_ϵ^2 is increasing. The results for the comparison of Bayesian method and PACE method are shown in Figure 6.

The smaller the MSE (ASE) value is, the better the method is. Hence, if the MSE (ASE) value of Bayesian method minus PACE method is less than zero, the Bayesian method is better; if the MSE (ASE) value of Bayesian method minus PACE method is greater than zero, the PACE method is better. Figures 5 and 6 show the 95% confidence interval for the MSE (ASE) value of Bayesian method minus PACE method. If the interval covers zero, the difference of two methods is not significant at the 95% level; if the interval does not cover zero, the difference of two methods is significant at the 95% level.

For the general performance under different n_i's and σ_ϵ^2's, panel (a) in Figures 5 and 6 indicate Bayesian method is significantly better than PACE method. For the estimations of baseline trend function under different n_i's and σ_ϵ^2's, panel (b) in Figures 5 and 6 show that PACE method is slightly better than Bayesian method when the data points are more sparse (i.e. $n_i = 1 - 5$) or more dense (i.e. $n_i = 15 - 30$). However, there is no significant difference for the estimation of the baseline trend function between the two methods.

From the results in simulation setting three Figure 5, Bayesian method is better for the estimations of all eigenfunctions and principal component scores. Especially, Bayesian method is significantly better for the estimation of the 1st eigenfunction when $n_i = 1 - 5$ (Panel (c)); is significantly better for the estimation of the 2nd eigenfunction when $n_i =$

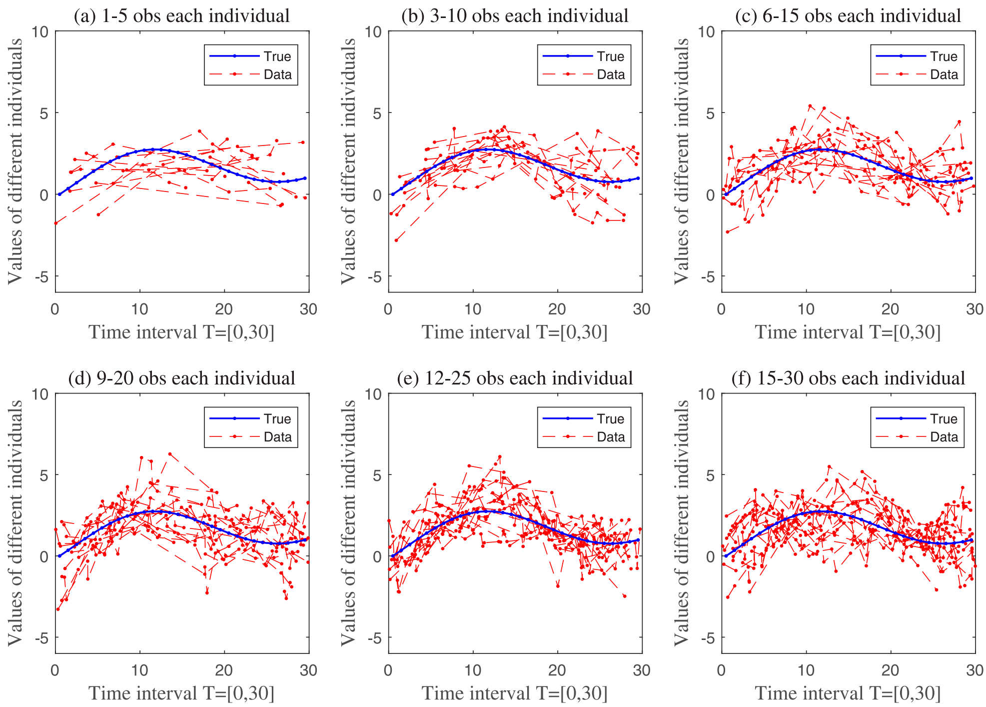

Figure 3. Six different simulation studies under $n_i = 1-5, 3-10, 6-15, 9-20, 12-15$ and $15-30$ with fixed $\sigma_\epsilon^2 = 0.7^2$. In each study, there are $\mathcal{N} = 25$ individuals (data sets) with n_i observations irregularly spaced on time interval $\mathcal{T} = [0, 30]$.

$1-5, 3-10, 6-15$ and $9-20$ (Panel (d)); and is significantly better for the estimations of the 3rd eigenfunction under any n_i's (Panel (e)). Bayesian method is significantly better for the estimation of the 1st principal component score when $n_i = 1-5$ and $3-10$ (Panel (f)); is significantly better for the estimation of the 2nd principal component score under any n_i's (Panel (g)); and is significantly better for the estimation of the 3rd principal component score for any n_i's except $n_i = 15-30$ (Panel (h)). In conclusion, when the data points are becoming sparser, the performance of Bayesian method is better.

From the results in simulation setting four Figure 6, Bayesian method is better for the estimations of all eigenfunctions and principal component scores. Especially, Bayesian method is significantly better for the estimations of the 2nd and 3rd eigenfunctions for all σ_ϵ^2's (Panel (d) and Panel (e)). Bayesian method is significantly better for the estimations of 1st principal component score when $\sigma_\epsilon^2 = 1.3^2$ and 1.5^2 (Panel (f)); and is significantly better for the estimations of 2nd and 3rd principal component scores (Panel (g) and Panel (h)). In conclusion, when the data points contain more noise, the performance of Bayesian method is better.

Overall, in simulation settings three and four, Bayesian method shows smaller MSE and ASE values in all estimations, indicating the proposed method is comparable with the PACE method. Especially, when the data set is more sparse or contains more noise, the proposed method has significantly better performance.

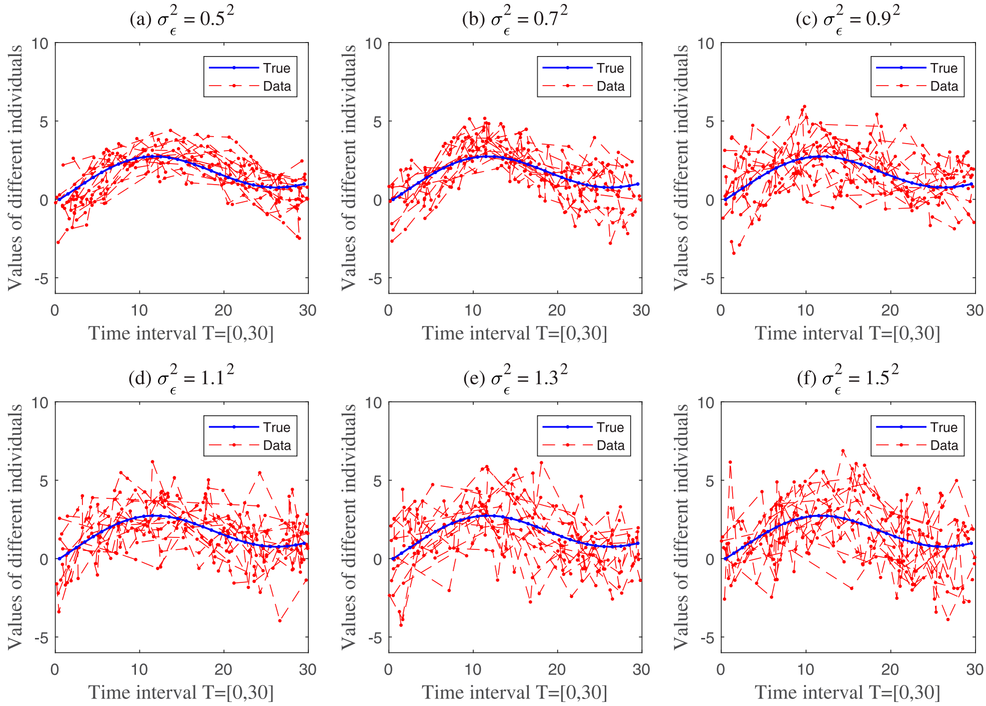

Figure 4. Six different simulation studies under $\sigma_\epsilon^2 = 0.5^2, 0.7^2, 0.9^2, 1.1^2, 1.3^2$ and 1.5^2. In each study, there are $\mathcal{N} = 25$ individuals (data sets) with $n_i = 9 - 20$ observations irregularly spaced on time interval $\mathcal{T} = [0, 30]$.

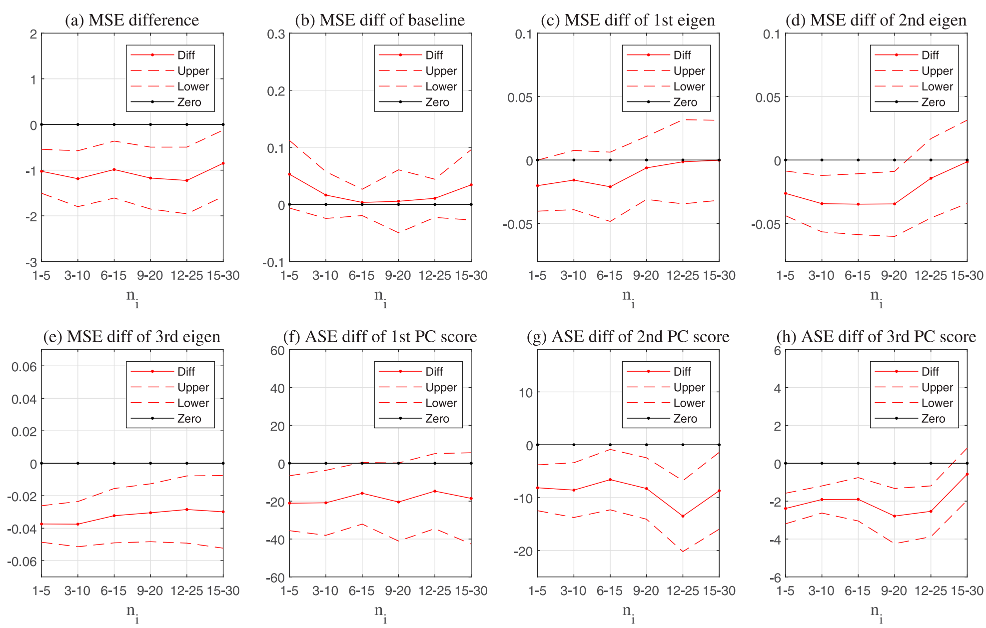

Figure 5. Comparison of the Bayesian method and the PACE method under different numbers of observations (i.e. n_i) in simulation. Each panel of graph shows values of the mean MSE (or ASE) of Bayesian minus the mean MSE (or ASE) of PACE and 95% confidence intervals of the mean differences.

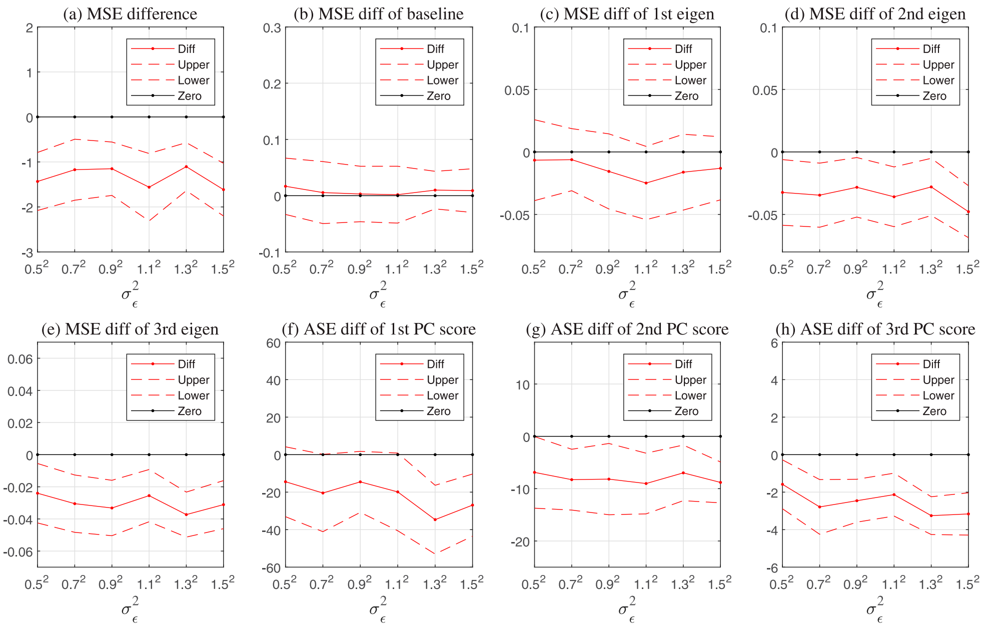

Figure 6. Comparison of the Bayesian method and the PACE method under different numbers of noises (i.e. σ_ϵ^2) in simulation. Each panel of graph shows values of the mean MSE (or ASE) of Bayesian minus the mean MSE (or ASE) of PACE and 95% confidence intervals of the mean differences.

3.1.3. Estimating convergence of MCMC

Convergence of the MCMC outputs for the different parameters was checked by multiple diagnostics [38]. Both Geweke's diagnostic and Heidelberger-Welch diagnostic showed stationarity of the chain after 30% of the initial iterations were discarded. The effective sample size (ESS) was at around 8,000 after discarding 4,000 'burn-in' iterations. Raftery-Lewis diagnostic confirmed this conclusion. Hence, 12,000 iterations were chosen with 30% 'burn-in' iterations in the simulation studies. Please see [38] for detailed explanations of these diagnostic tools.

3.2. Real data analysis

The proposed method is illustrated to the analysis of body mass index (BMI) data by gender and ethnicity. This study is motivated by the fact that the BMI data by gender and ethnicity are sparse and irregularly spaced with missing observations. Hence, it is necessary to consider them as functional data [57].

3.2.1. Background of the body mass index data

BMI is a measure of relative size based on the mass and height of an individual. If the weight is in kilograms and the height in meters, BMI is measured by the ratio of weight to squared height, i.e. kg/m^2. People at different ages have different risk of overweight, where adult overweight is defined as a BMI $\geq 25\,\text{kg/m}^2$. So the data have binary outcomes for each individual: overweight and not overweight. The dramatic increase in the prevalence of overweight in the US population during the past few decades has drawn much research

Table 3. Numbers of individuals, related unique observations, and missing observations under the matrix of age × period (i.e. 68 × 17) by gender and ethnicity.

	Individuals			Unique Observations			Missing Observations		
	Females	Males	All	Females	Males	All	Females	Males	All
All	3179	2479	5658	1054	974	1132	102	182	24
Asians	112	105	217	102	101	185	1054	1055	971
Blacks	459	307	766	368	265	542	788	891	614
Hispanics	553	406	959	410	325	587	746	831	569
Whites	2055	1661	3716	942	854	1097	214	302	59

attention [34,35]. The purpose of the study is to investigate the prevalence trajectories of BMI for different age groups during a specific period (i.e. from 1997 to 2013 in this article), which is called age-period analysis.

The National Health Interview Survey (NHIS) provides researchers with national representative BMI information for US adults. In this article, the object of the analysis is to verify the performance of the proposed Bayesian model. Therefore, a random subset of 6,000 individuals was selected from the original NHIS data where the data were pooled from 1997 to 2013. After deleting cases with invalid weight and height information, the analytic sample contains 5,658 individuals with ages ranging from 0 to 67 years old, where each individual is either overweight or not overweight.

The age-period analysis is performed for binary trajectories $\{S_i(t_{ij}),\ t_{ij} \in \mathcal{T}, j = 1, \ldots, n_i\}$ with $S_i(t_{ij}) = 1$ if overweight and $S_i(t_{ij}) = 0$ otherwise. The index of age is denoted as $i = 1, \ldots, 68$ (i.e. from 0 to 67 years old); and the index of period is denoted as $j = 1, \ldots, n_i$ with $n_i = 1 - 17$ irregularly spaced on $\mathcal{T} = [1, 17]$ (i.e. from 1997 to 2013). Table 3 shows numbers of individuals, related unique observations and missing observations under the matrices of age × period (i.e. 68 × 17) by gender and ethnicity. Within the 5,658 individuals in the study, there are 112 female Asians, 105 male Asians; 459 female Blacks, 307 male Blacks; 553 female Hispanics, 406 male Hispanics; and 2,055 female Whites, 1,661 male Whites. The sum of the number of unique observations and the number of missing observations is 1,156 under age × period (i.e. 68 × 17). When number of individuals decreases, the data set becomes more sparse with more missing observations.

In Table 3, the BMI data stratified by gender and ethnicity are typically sparse within the subgroups. Although pooling subgroups together will reduce the sparseness of the data, there are reasons that researchers are cautious about pooling gender and ethnic groups together. Previous research in sociology has established significant differences in the prevalence of overweight and obesity across gender and ethnic groups [34]. Weight status is closely linked to gender, diet, culture and lifestyles. Different ethnic groups are influenced by different culture and ethnic life styles. Hence, researchers in sociology are reluctant to pool gender and ethnic groups together and assume similar trends across these groups. In the study, the focus is on analyzing the data stratified by gender and ethnicity. The major merit of the proposed model is its capacity of analyzing sparse and irregularly spaced data.

3.2.2. *Model fitting*

For each ethnicity group by gender, the FPCA analysis was performed independently using the Bayesian mixed-effects model. The RJ-MCMC ran until the convergence, where

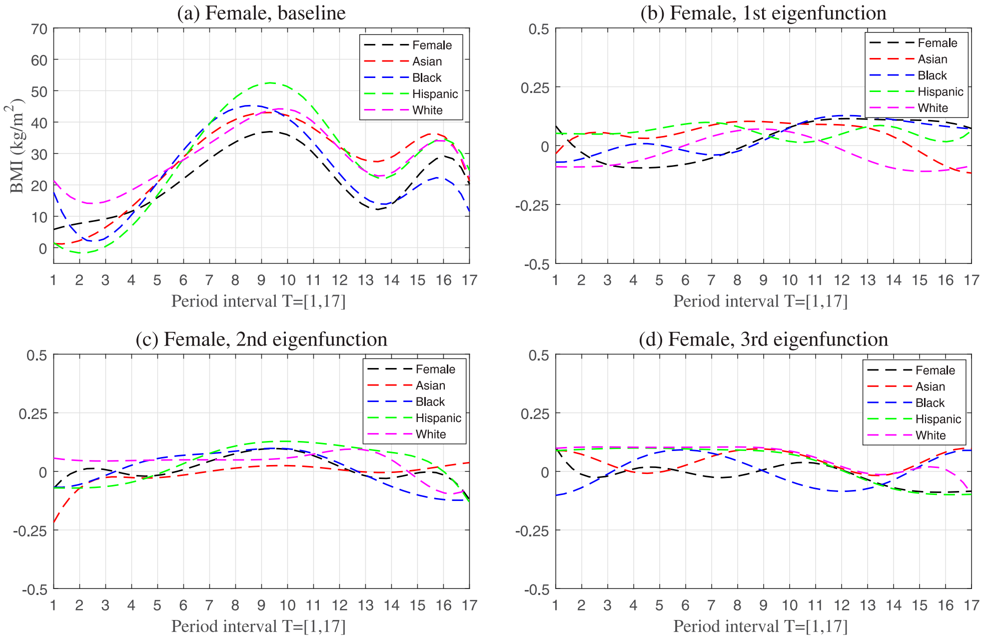

Figure 7. Curves for females over period interval $\mathcal{T} = [1, 17]$, corresponding to years 1997-2013. Panel (a) is the baseline BMI trend curves (i.e. $\mathrm{kg/m^2}$ values) by ethnicity. Panels (b), (c) and (d) are first three eigenfunctions over period effect by ethnicity. The eigenfunctions correspond to the different effects shifting from the baseline trend curve, and reflect variations about the baseline trend curve.

only one principal component was picked up in every independent analysis procedure. In age-period analysis, the curves are decomposed as sum of a fixed baseline BMI trend curve during period interval $\mathcal{T} = [1, 17]$ (i.e. from 1997 to 2013) and some random deviations from the baseline curve. The deviations are subsequently summarized by a few smoothed eigenfunctions extracted from the period trend subtracted BMI curve. Figures 7 and 8 show the baseline BMI trend curves (i.e. $\mathrm{kg/m^2}$ values) and three eigenfunctions by gender and ethnicity. Looking at the 2nd and 3rd eigenfunctions, since the amplitude of each curve is relatively lower, both eigenfunctions are corrections of the 1st eigenfunction.

In Figures 7 and 8, panel (a) shows the badeline BMI trend curves (i.e. $\mathrm{kg/m^2}$ values) by gender and ethnicity. Both females and males have generally similar trends, the peaks are reached at period 9 (i.e. 2005) and period 16 (i.e. 2012). Among the ethnic groups, female Hispanics and male Asians have the highest BMI at period 9 (i.e. 2005). In general, the trends in females are less substantial and more diversity; and the trends in males are more substantial and concentrated.

There are three principal components in the procedure, panels (b), (c) and (d) in Figures 7 and 8 reveal the three eigenfunctions during period interval $\mathcal{T} = [1, 17]$ (i.e. from 1997 to 2013) by gender and ethnicity. The eigenfunctions correspond to the different effects shifting from the baseline trend, and reflect variations about the trend curve over $\mathcal{T} = [1, 17]$ (i.e. from 1997 to 2013). Since the 1st principal component is the most important, the 1st eigenfunction essentially yields a summary statistics of the overweight trend. For females, overweight generally increases starting from period 8 (i.e. 2004), especially in Blacks and

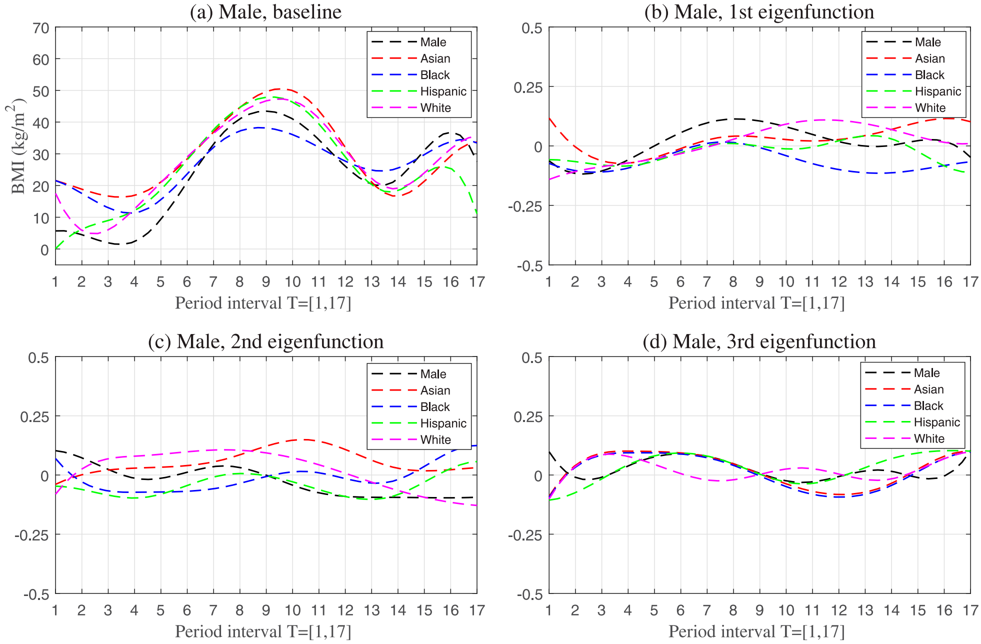

Figure 8. Curves for males over period interval $\mathcal{T} = [1, 17]$, corresponding to years 1997-2013. Panel (a) is the baseline BMI trend curves (i.e. kg/m^2 values) by ethnicity. Panels (b), (c) and (d) are first three eigenfunctions over period effect by ethnicity. The eigenfunctions correspond to the different effects shifting from the baseline trend curve, and reflect variations about the baseline trend curve.

Hispanics. For males, overweight generally increases starting from period 5 (i.e. 2001), especially in Asians and Whites. The 2nd and 3rd eigenfunctions show relatively lower waves during the period, which explain less variances in the BMI data and can be viewed as corrections of the 1st eigenfunction. To make a comparison, BMI data analysis was performed by the PACE-GRM method too. The results from the PACE-GRM method are very similar to those from the proposed Bayesian method. Especially, the shapes of those curves from the two methods are very close. This property is consistent with the conclusions from the simulation studies (Figure 2 and Table 2).

4. Conclusion remarks

In this article, a Bayesian method in FPCA is proposed, where Bayesian inference is based on MCMC. This method works very well for both continuous and binary responses which are sparse and irregularly spaced. This approach relies on appropriate prior distributions for unknown parameters, and the use of Gibbs sampling performs quite effectively in practice for both simulation studies and real BMI data analysis. Moreover, this method is capable of automatically determining the number of principal components and estimating the other unknown parameters simultaneously. The simulation settings demonstrate that the proposed method is comparable with the FPCA methods in frequentist inference. In the BMI data analysis, the proposed method adds a great degree of flexibility in the age-period analysis using a nonparametric model structure. The method is especially suitable for the data with sparse and irregularly spaced measurements. The use of Bayesian analysis increases flexibility of the estimates, improves the statistical and numerical spatiality of the estimations. All algorithms are implemented by Matlab.

In FPCA, the trajectories were represented directly through the Karhunen-Loève expansion. This transformation is an eigenvector transform for the trajectories, determining the eigenfunctions from the sparse data. However, most of the researchers did not investigate the asymptotic properties of estimations, such as the estimated covariance structure, eigenvalues and eigenfunctions, especially for the sparse situation [55]. Instead, bootstrap was used to constructed for the pointwise confidence intervals of individual curves [22]. In Bayesian statistics, there is no such a concern as the statistical inference is based on credible interval of the simulated samples.

Functional analysis of variance (ANOVA) is a regression problem where some of the covariates are functional. The most common data-driven device in FDA is FPCA [29]. In the study of functional ANOVA in FPCA, the complicated frequentist approach has to rely on bootstrap [50] or permutation [54] for hypothesis tests. In the test procedures on the mean function under generalized likelihood ratio (GLR) test statistics, there are three versions of GLR tests based on the marginal likelihood, conditional likelihood, and working independence (WI), respectively. Tang *et al.* [50] approximated the null distribution using bootstrap and bandwidth selection. Xu *et al.* [54] proposed a permutation procedure to evaluate the null distribution of the test statistics. With the Bayesian approach in the proposed method, statistical inference could be directly derived from the results of MCMC. Hence, it is interesting to compare the Bayesian method with bootstrap and/or permutation approaches for hypothesis tests in functional ANOVA. This research topic will be explored in future work.

Recently, FDA has been used in electricity market. Since the electricity data are usually rough [3], functional methods in time series are very useful for electricity demand forecast [3,4,23,40,41]. Some researchers proposed a functional autoregressive model of order p (FAR(p)) under the Hilbert space to predict sale and purchase curves of electricity prices [23,40]. In order to deal with the non-invertibility of the covariance operator in FAR(p), two approaches are typically used: truncation and penalization. In the truncation approach, FPCA is performed, where data are projected to a subspace of finite dimension spanned by the eigenfunctions of the covariance operator, with the number of the eigenvalues determined by cross-validation [40]. In the penalization approach, the generalized penalty is involved on the basis coefficients, leading to a ridge regression structure, where statistical inference is based on bootstrap [22]. In the literature, it seems that the researchers prefer the truncation approach through FPCA [23,40]. In FAR(p) under the Hilbert space, the proposed Bayesian FPCA method by RJ-MCMC can be used as it considers both truncation and penalization approaches. This research topic will be explored in future work.

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Appendices

Appendix A. Technical details of the method

A.1 The Langevin-Bingham matrix variate distribution

By [10,18], Stiefel manifold $V_{p,q}$ in q-dimensional real space R^q is represented as the set of $(q \times p), q \geq p$ random matrices X such that $X^TX = I_p$, where I_p is a $p \times p$ identity matrix. Define $V_{p,q} = \{X(q \times p); X^TX = I_p\}$, the $\mathrm{vol}(V_{p,q})$ is a surface area (referred to as the volume) of the Stiefel manifold $V_{p,q}$, i.e.

$$\mathrm{vol}(V_{p,q}) = \int_{V_{p,q}} X^T\,\mathrm{d}X = \frac{2^p \pi^{\frac{pq}{2}}}{\Gamma_p\left(\frac{q}{2}\right)}, \tag{A1}$$

where $\Gamma_p(\cdot)$ is a multivariate Gamma function with $\Gamma_p(u) = \pi^{\frac{p(p-1)}{4}} \prod_{k=1}^{p} \Gamma(u - \frac{k-1}{2})$, and $\Gamma(\cdot)$ is the Gamma function with $\Gamma(u) = \int_0^\infty t^{u-1}\,\mathrm{e}^{-t}\,\mathrm{d}t$. The differential form $X^T dX$ represents invariant measure on $V_{p,q}$, which can be normalized by $\mathrm{vol}(V_{p,q})$ to provide a probability distribution on $V_{p,q}$. Hence the unit invariant measure on $V_{p,q}$ is denoted by

$$[\mathrm{d}X] = \frac{1}{\mathrm{vol}(V_{p,q})}[X^T\,\mathrm{d}X], \tag{A2}$$

which can be a uniform probability distribution on the Stiefel manifold. Many probability distributions on $V_{p,q}$ have exponential family forms. Define $\mathrm{etr}(\mathbb{S}) = \exp(\mathrm{tr}\mathbb{S})$ where $\mathrm{tr}\mathbb{S}$ is the trace of the matrix $\mathbb{S}$ in the formula, a random matrix X has Langevin (or von Mises-Fisher) $L(q, p; \mathbb{F})$ distribution with parameter $\mathbb{F}$ if its density function is

$$a(\mathbb{F})\mathrm{etr}(\mathbb{F}^TX)[\mathrm{d}X], \tag{A3}$$

where $\mathbb{F}$ is a $(q \times p)$ matrix and $a(\mathbb{F})$ is a normalizing constant. A random matrix X has Bingham $B(q, p; \mathbb{A}, \mathbb{B})$ distribution with parameters $\mathbb{A}, \mathbb{B}$ if its density function is

$$b(\mathbb{A}, \mathbb{B})\mathrm{etr}(\mathbb{A}X^T\mathbb{B}X)[\mathrm{d}X], \tag{A4}$$

where $\mathbb{B}$ is a $q \times q$ symmetric matrix, $\mathbb{A}$ is a $p \times p$ symmetric matrix, and $b(\mathbb{A}, \mathbb{B})$ is a normalizing constant. From above, the general Langevin-Bingham matrix variate distribution has the density function proportional to

$$\mathrm{etr}(\mathbb{F}^TX + \mathbb{A}X^T\mathbb{B}X)[\mathrm{d}X]. \tag{A5}$$

A.2 The conditional posterior distributions

Conditional posterior distributions required for the MCMC algorithms are presented here.

A.2.1 Estimator of the liability Y under binary outcomes

Under normal distribution $N(\mu, \sigma^2)$, let $\phi(\frac{y_i-\mu}{\sigma})$ be the probability density function, and $\Phi(\frac{y_i-\mu}{\sigma})$ be the cumulative distribution function, the truncated normal distribution is

$$\mathrm{TNormal}(\mu, \sigma^2, a, b) = \frac{\frac{1}{\sigma}\Phi(\frac{y_i-\mu}{\sigma})}{\Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})}, \tag{A6}$$

where $\Phi(\frac{b-\mu}{\sigma}) = 1$ if $b = \infty$ and $\Phi(\frac{a-\mu}{\sigma}) = 0$ if $a = -\infty$. The conditional posterior distribution of the liability $Y_i, i = 1, \ldots, \mathcal{N}$, follows a truncated multivariate normal distribution. Depending on the binary value S_i, Y_i is generated by truncated multivariate normal distributions

$$f(Y_i|S_i, S_i = \mathbf{1}) = \mathrm{TNormal}(A_i\theta + B_i\Theta_\psi E_\xi \Theta_{\xi i}^T, \mathbf{1}, \mathbf{0}, \infty), \tag{A7}$$

and

$$f(Y_i|S_i, S_i = \mathbf{0}) = \mathrm{TNormal}(A_i\theta + B_i\Theta_\psi E_\xi \Theta_{\xi i}^T . \mathbf{1}, -\infty, \mathbf{0}), \tag{A8}$$

where $\mathbf{0}$ and $\mathbf{1}$ are $n_i \times 1$ vectors.

A.2.2 Estimators under continuous outcomes

A.2.2.1 Estimator of $\boldsymbol{\Theta_\psi}$.

Let $\theta_{\psi l} = \Theta_{\psi[\cdot,l]}, l = 1, \ldots, p$, the likelihood function of $\theta_{\psi l}$ is

$$\begin{aligned}
&f(\mathbf{Y}|\theta_{\psi l}, \theta_{\xi l}, E_{\xi[l,l]}, \sigma_\varepsilon^2, \theta, h_\psi) \\
&= \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}} \| Y_i - A_i\theta - B_i\Theta_{\psi[\cdot,-l]}E_{\xi[-l,-l]}\Theta_{\xi i[\cdot,-l]}^T - B_i\Theta_{\psi[\cdot,l]}e_l\theta_{\xi il} \|^2 + \frac{1}{2}h_\psi\theta_{\psi l}^T\mathcal{J}_\mathcal{B}\theta_{\psi l}\right\}\right] \\
&\propto \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}}(-2)e_l(B_i\theta_{\psi l})^T(Y_i - A_i\theta - B_i\Theta_{\psi[\cdot,-l]}E_{\xi[-l,-l]}\Theta_{\xi i[\cdot,-l]}^T)\theta_{\xi il} + \frac{1}{2}h_\psi\theta_{\psi l}^T\mathcal{J}_\mathcal{B}\theta_{\psi l}\right\}\right] \\
&\propto \exp\left(\frac{1}{\sigma_\varepsilon^2}e_l\theta_{\psi l}^TZ\theta_{\xi l} - \frac{1}{4\sigma_\varepsilon^2}h_\psi\theta_{\psi l}^T\mathcal{J}_\mathcal{B}\theta_{\psi l}\right),
\end{aligned} \tag{A9}$$

where

$$Z_i = B_i^T(Y_i - A_i\theta - B_i\Theta_{\psi[\cdot,-l]}E_{\xi[-l,-l]}\Theta_{\xi i[\cdot,-l]}^T), \tag{A10}$$

and $Z = (Z_1, \ldots, Z_\mathcal{N})$ is a $q \times \mathcal{N}$ matrix. Then the conditional posterior distribution is

$$\begin{aligned}
&f(\theta_{\psi l}|\Theta_{\psi[\cdot,-l]}), \text{others}) \\
&\propto \exp\left(\frac{1}{\sigma_\varepsilon^2}e_l\theta_{\psi l}^TZ\theta_{\xi l} - \frac{1}{4\sigma_\varepsilon^2}h_\psi\theta_{\psi l}^T\mathcal{J}_\mathcal{B}\theta_{\psi l}\right)\mathbb{1}(\theta_{\psi l} \in V_{1,q}|\Theta_{\psi[\cdot,-l]}).
\end{aligned} \tag{A11}$$

Conditionally upon $\Theta_{\psi[\cdot,-l]}$, let $\theta_{\psi l} = M_l \nu_l$ [5,7], the conditional posterior distribution for ν_l is

$$\begin{aligned} & f(\nu_l|\Theta_{\psi[\cdot,-l]}, \text{others}) \\ & \propto \exp\left\{\frac{1}{\sigma_\varepsilon^2} e_l (M_l\nu_l)^T Z\theta_{\xi l} - \frac{1}{4\sigma_\varepsilon^2} h_\psi (M_l \nu_l)^T \mathcal{J}_{\mathcal{B}} M_l \nu_l\right\} \\ & \propto \exp\left(\frac{1}{\sigma_\varepsilon^2} e_l \theta_{\xi l}^T Z^T M_l \nu_l - \frac{1}{4\sigma_\varepsilon^2} h_\psi \nu_l^T M_l^T \mathcal{J}_{\mathcal{B}} M_l \nu_l\right) \\ & \propto \exp(\mathbb{F}^T \nu_l + \mathbb{A} \nu_l^T \mathbb{B} \nu_l). \end{aligned} \tag{A12}$$

This is a Langevin-Bingham matrix variate distribution [10,18] with parameters $\mathbb{F} = \frac{1}{\sigma_\varepsilon^2} e_l M_l^T Z\theta_{\xi l}$, $\mathbb{A} = -\frac{1}{4\sigma_\varepsilon^2} h_\psi$, and $\mathbb{B} = M_l^T \mathcal{J}_{\mathcal{B}} M_l$. Columns of Θ_ψ (i.e. $\theta_{\psi l}$, $l = 1, \ldots, p$) can be iteratively sampled conditionally upon the others (i.e. $\Theta_{\psi[\cdot,-l]}$) by drawing sample ν_l from a Langevin-Bingham distribution and setting

$$\theta_{\psi l} = M_l \nu_l. \tag{A13}$$

A.2.2.2 Estimator of $\boldsymbol{\Theta}_\xi$.

Let $\theta_{\xi l} = \Theta_{\xi[\cdot,l]}$, $l = 1, \ldots, p$, the likelihood function of $\theta_{\xi l}$ is

$$\begin{aligned} & f(\mathbf{Y}|\theta_{\psi l}, \theta_{\xi l}, E_{\xi[l,l]}, \sigma_\varepsilon^2, \theta) \\ & = \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}} \| Y_i - A_i\theta - B_i \Theta_{\psi[\cdot,-l]} E_{\xi[-l]} \Theta_{\xi i[\cdot,-l]}^T - B_i \theta_{\psi l} e_l \theta_{\xi il} \|^2\right\}\right] \\ & \propto \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}} (-2) e_l (B_i\theta_{\psi l})^T (Y_i - A_i\theta - B_i\Theta_{\psi[\cdot,-l]} E_{\xi[-l]} \Theta_{\xi i[\cdot,-l]}^T)\theta_{\xi il}\right\}\right] \\ & \propto \exp\left(\frac{1}{\sigma_\varepsilon^2} e_l \theta_{\psi l}^T Z\theta_{\xi l}\right), \end{aligned} \tag{A14}$$

where

$$Z_i = B_i^T (Y_i - A_i\theta - B_i\Theta_{\psi[\cdot,-l]} E_{\xi[-l,-l]} \Theta_{\xi i[\cdot,-l]}^T), \tag{A15}$$

and $Z = (Z_1, \ldots, Z_{\mathcal{N}})$ is a $q \times \mathcal{N}$ matrix. Then the conditional posterior distribution is

$$\begin{aligned} & f(\theta_{\xi l}|\Theta_{\xi[\cdot,-l]}, \text{others}) \\ & \propto \exp\left(\frac{1}{\sigma_\varepsilon^2} e_l \theta_{\psi l}^T Z\theta_{\xi l}\right) \mathbb{1}(\theta_{\xi l} \in V_{1,\mathcal{N}} | \Theta_{\xi[\cdot,-l]}). \end{aligned} \tag{A16}$$

Conditionally upon $\Theta_{\xi[\cdot,-l]}$, let $\theta_{\xi l} = M_{\xi l} \nu_{\xi l}$, the conditional posterior distribution for $\nu_{\xi l}$ is

$$\begin{aligned} & f(\nu_{\xi l}|\Theta_{\xi[\cdot,-l]}, \text{others}) \\ & \propto \exp\left(\frac{1}{\sigma_\varepsilon^2} e_l \theta_{\psi l}^T Z M_{\xi l} \nu_{\xi l}\right) \\ & \propto \exp(\mathbb{F}_\xi^T \nu_{\xi l}), \end{aligned} \tag{A17}$$

where $\mathbb{F}_\xi = e_l M_{\xi l}^T Z^T \theta_{\psi l}$ is a $(\mathcal{N} - p + 1) \times 1$ matrix of coefficients in a von Mises-Fisher distribution [10,18]. Columns of Θ_ξ (i.e. $\theta_{\xi l}$, $l = 1, \ldots, p$) can be iteratively sampled conditionally upon the others (i.e. $\Theta_{\xi[\cdot,-l]}$) by drawing sample ν_l from a von Mises–Fisher distribution and setting

$$\theta_{\xi l} = M_{\xi l} \nu_{\xi l}. \tag{A18}$$

A.2.2.3 Estimation of E_ξ. Let $e_l = E_{\xi[l,l]}$, $l = 1, \ldots, p$, the likelihood function of e_l is

$$f(\mathbf{Y}|\text{others}) \propto \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{(-2)\sum_{i=1}^{\mathcal{N}} e_l(B_i\theta_{\psi l})^T(Y_i - A_i\theta - B_i\Theta_{\psi[\cdot,-l]}E_{\xi[-l]}\Theta^T_{\xi i[\cdot,l]})\theta_{\xi il}\right\}\right]$$

$$\exp\left\{-\frac{1}{2\sigma_\varepsilon^2}\sum_{i=1}^{\mathcal{N}}\|B_i\theta_{\psi l}e_l\theta_{\xi il}\|^2\right\}$$

$$\propto \exp\left(\frac{1}{\sigma_\varepsilon^2}\theta_{\psi l}^T Z\theta_{\xi l}e_l + \frac{1}{2\sigma_\varepsilon^2}e_l^2\right), \tag{A19}$$

where

$$Z_i = B_i^T(Y_i - A_i\theta - B_i\Theta_{\psi[\cdot,-l]}E_{\xi[-l,-l]}\Theta^T_{\xi i[\cdot,-l]}), \tag{A20}$$

and $Z = (Z_1, \ldots, Z_{\mathcal{N}})$ is a $q \times \mathcal{N}$ matrix. The conditional posterior distribution is

$$f(e_l|\text{others}) = f(\mathbf{Y}|\text{others})f(e_l)$$

$$\propto \exp\left(\frac{1}{\sigma_\varepsilon^2}\theta_{\psi l}^T Z\theta_{\xi l}e_l - \frac{1}{2\sigma_\varepsilon^2}e_l^2\right)\exp\left\{-\frac{1}{2\sigma_e^2}(e_l - \mu_e)^2\right\}$$

$$\propto \exp\left(-\frac{1}{2}A_e e_l^2 + B_e e_l\right), \tag{A21}$$

where $A_e = \frac{1}{\sigma_e^2} + \frac{1}{\sigma_\varepsilon^2}$, $B_e = \frac{\mu_e}{\sigma_e^2} + \frac{\theta_{\psi l}^T Z\theta_{\xi l}}{\sigma_\varepsilon^2}$. This is a normal distribution with $E(e_l|\text{others}) = \frac{B_e}{A_e}$ and $Var(e_l|\text{others}) = \frac{1}{A_e}$.

A.2.2.4 Estimator of σ_ε^2. The likelihood function for a function of σ_ε^2 is

$$f(\mathbf{Y}|\xi_i, \sigma_\varepsilon^2, \theta, \Theta_\psi, h_\mu, h_\psi)$$

$$\propto \left(\frac{1}{\sigma_\varepsilon^2}\right)^{\sum_{i=1}^{\mathcal{N}}\frac{n_i}{2}}\exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}}\| Y_i - A_i\theta - B_i\Theta_\psi E_\xi\Theta_{\xi i}^T\|^2\right\}\right]$$

$$\exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\frac{1}{2}h_\mu\theta_\mu^T\mathcal{J}_{\mathcal{B}}\theta_\mu + \frac{1}{2}h_\psi\sum_{l=1}^{p}\theta_{\psi l}^T\mathcal{J}_{\mathcal{B}}\theta_{\psi l}\right\}\right]. \tag{A22}$$

The conditional posterior distribution is

$$f\left(\frac{1}{\sigma_\varepsilon^2}|\text{others}\right) = f(\mathbf{Y}|\Theta_\psi, \Theta_\xi, E_\xi, \sigma_\varepsilon^2, \theta, h_\mu, h_\psi)f(\frac{1}{\sigma_\varepsilon^2})$$

$$\propto \left(\frac{1}{\sigma_\varepsilon^2}\right)^{\sum_{i=1}^{\mathcal{N}}\frac{n_i}{2}}\exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}}\| Y_i - A_i\theta - B_i\Theta_\psi E_\xi\Theta_{\xi i}^T\|^2\right\}\right]$$

$$\exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\frac{1}{2}h_\mu\theta_\mu^T\mathcal{J}_{\mathcal{B}}\theta_\mu + \frac{1}{2}h_\psi\sum_{l=1}^{p}\theta_{\psi l}^T\mathcal{J}_{\mathcal{B}}\theta_{\psi l}\right\}\right]\left(\frac{1}{\sigma_\varepsilon^2}\right)^{c_\varepsilon - 1}\exp\left(-d_\varepsilon\frac{1}{\sigma_\varepsilon^2}\right). \tag{A23}$$

Hence,

$$f\left(\frac{1}{\sigma_\varepsilon^2}|\text{others}\right) \sim \text{Gamma}$$

$$\left(\sum_{i=1}^{\mathcal{N}}\frac{n_i}{2} + c_\varepsilon, \frac{1}{2}\sum_{i=1}^{\mathcal{N}}\| Y_i - A_i\theta - B_i\Theta_\psi E_\xi\Theta_{\xi i}^T\|^2 + \frac{1}{4}h_\mu\theta_\mu^T\mathcal{J}_{\mathcal{B}}\theta_\mu + \frac{1}{4}h_\psi\sum_{l=1}^{p}\theta_{\psi l}^T\mathcal{J}_{\mathcal{B}}\theta_{\psi l} + d_\varepsilon\right). \tag{A24}$$

A.2.2.5 ***Estimator of $\boldsymbol{\theta}$.*** The likelihood function of θ is

$$
\begin{aligned}
&f(\mathbf{Y}|\Theta_\psi,\Theta_\xi,E_\xi,\sigma_\varepsilon^2,\theta,h_\mu,h_\psi)\\
&\quad\propto \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}}\parallel Y_i-A_i\theta-B_i\Theta_\psi E_\xi\Theta_{\xi i}^T\parallel^2+\frac{1}{2}h_\mu\theta^T\mathcal{J}_\mathcal{A}\theta\right\}\right]\\
&\quad\propto \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}}\theta^TA_i^TA_i\theta-2\sum_{i=1}^{\mathcal{N}}\theta^TA_i^T(Y_i-B_i\Theta_\psi E_\xi\Theta_{\xi i}^T)+\frac{1}{2}h_\mu\theta^T\mathcal{J}_\mathcal{A}\theta\right\}\right]. \qquad \text{(A25)}
\end{aligned}
$$

The conditional posterior is

$$
\begin{aligned}
&f(\theta|\text{others})=f(\mathbf{Y}|\Theta_\psi,\Theta_\xi,E_\xi,\sigma_\varepsilon^2,\theta,h_\mu,h_\psi)f(\theta|h_\mu,\sigma_\varepsilon^2)\\
&\quad\propto \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}}\theta^TA_i^TA_i\theta+h_\mu\theta^T\mathcal{J}_\mathcal{A}\theta\right\}+\frac{1}{\sigma_\varepsilon^2}\left\{\sum_{i=1}^{\mathcal{N}}\theta^TA_i^T(Y_i-B_i\Theta_\psi E_\xi\Theta_{\xi i}^T)\right\}\right]\\
&\quad=\exp\left(-\frac{1}{2}\theta^TA_\mu\theta+\theta^Tb_\mu\right), \qquad \text{(A26)}
\end{aligned}
$$

with $A_\mu=\frac{1}{\sigma_\varepsilon^2}(\sum_{i=1}^{\mathcal{N}}A_i^TA_i+h_\mu\mathcal{J}_\mathcal{A})$ and $b_\mu=\frac{1}{\sigma_\varepsilon^2}\sum_{i=1}^{\mathcal{N}}A_i^T(Y_i-B_i\Theta_\psi E_\xi\Theta_{\xi i}^T)$. This is a multivariate normal distribution

$$
\begin{aligned}
&f(\theta|\text{others})\sim\text{Normal}(E(\theta|\text{others}),\text{Cov}(\theta|\text{others})),\quad \text{where}\\
&E(\theta|\text{others})=(A_\mu)^{-1}b_\mu=\left(\sum_{i=1}^{\mathcal{N}}A_i^TA_i+h_\mu\mathcal{J}_\mathcal{A}\right)^{-1}\sum_{i=1}^{\mathcal{N}}A_i^T(Y_i-B_i\Theta_\psi E_\xi\Theta_{\xi i}^T),\\
&\text{Cov}(\theta|\text{others})=(A_\mu)^{-1}=\left(\sum_{i=1}^{\mathcal{N}}A_i^TA_i+h_\mu\mathcal{J}_\mathcal{A}\right)^{-1}. \qquad \text{(A27)}
\end{aligned}
$$

A.2.2.6 ***Estimator of $\boldsymbol{h_\mu,h_\psi}$.*** The likelihood function of h_μ is

$$f(\theta_\mu|h_\mu)\propto(h_\mu)^{\frac{q-r}{2}}\exp\left(-\frac{h_\mu\theta_\mu^T\mathcal{J}_\mathcal{B}\theta_\mu}{2}\right), \qquad \text{(A28)}$$

where r is the order of the B-splines and q is the number of the basis functions [2,46,49] The conditional posterior is

$$
\begin{aligned}
f(h_\mu|\theta_\mu)&\propto f(\theta_\mu|h_\mu)f(h_\mu)\\
&\propto(h_\mu)^{\frac{q-r}{2}}\exp\left(-\frac{h_\mu\theta_\mu^T\mathcal{J}_\mathcal{B}\theta_\mu}{2}\right)(h_\mu)^{c_\mu-1}\exp(-d_\mu h_\mu)\\
&\sim\text{Gamma}\left(\frac{q-r}{2}+c_\mu,\frac{\theta_\mu^T\mathcal{J}_\mathcal{B}\theta_\mu}{2}+d_\mu\right). \qquad \text{(A29)}
\end{aligned}
$$

Since the likelihood $f(\Theta_\psi|h_\psi)=f(\Theta_\psi)$ is a uniform distribution, the conditional posterior distribution is $f(h_\psi|\Theta_\psi)=f(h_\psi)$. By Equation (27),

$$f(h_\psi)\sim\text{Gamma}(c_\psi,d_\psi). \qquad \text{(A30)}$$

A.2.2.7 ***Estimator of $\boldsymbol{\mu_e}$.*** The likelihood function of e_l is

$$f(e_l|\mu_e,\sigma_e^2)\propto\exp\left\{-\frac{1}{2\sigma_e^2}\sum_{l=1}^{p}(e_l-\mu_e)^2\right\}. \qquad \text{(A31)}$$

Considering μ_e is normally distributed, i.e.

$$f(\mu_e) \propto \exp\left\{-\frac{1}{2\nu_{eo}^2}(\mu_e - \mu_{eo})^2\right\}, \tag{A32}$$

the conditional posterior of μ_e is

$$f(\mu_e|\text{others}) \propto \exp\left\{-\frac{1}{2}\left(\frac{1}{\nu_{eo}^2} + \frac{p}{\sigma_e^2}\right)\mu_e^2 + \left(\frac{\mu_{eo}}{\nu_{eo}^2} + \frac{1}{\sigma_e^2}\sum_{l=1}^{p} e_l\right)\mu_e\right\}. \tag{A33}$$

This is a normal distribution with

$$Var(\mu_e|\text{others}) = \left(\frac{1}{\nu_{eo}^2} + \frac{p}{\sigma_e^2}\right)^{-1} \tag{A34}$$

$$E(\mu_e|\text{others}) = \left(\frac{1}{\nu_{eo}^2} + \frac{p}{\sigma_e^2}\right)^{-1}\left(\frac{1}{\nu_{eo}^2}\mu_{eo} + \frac{1}{\sigma_e^2}\sum_{l=1}^{p} e_l\right) \tag{A35}$$

A.2.2.8 Estimator of σ_e^2. Considering $f(\frac{1}{\sigma_e^2}) \sim \text{Gamma}(c_e, d_e)$, the conditional posterior distribution of $f(\frac{1}{\sigma_e^2})$ is

$$\begin{aligned} f(\frac{1}{\sigma_e^2}|\text{others}) &\propto f(e_l|\mu_e, \sigma_e^2) f(\frac{1}{\sigma_e^2}) \\ &\propto \left(\frac{1}{\sigma_e^2}\right)^{p/2} \exp\left\{-\frac{1}{2\sigma_e^2}\sum_{l=1}^{p}(e_l - \mu_e)^2\right\}\left(\frac{1}{\sigma_e^2}\right)^{c_e - 1}\exp\left(\frac{-d_e}{\sigma_e^2}\right) \\ &\propto \left(\frac{1}{\sigma_e^2}\right)^{p/2 + c_e - 1}\exp\left[-\frac{1}{\sigma_e^2}\left\{\frac{\sum_{l=1}^{p}(e_l - \mu_e)^2}{2} + d_e\right\}\right], \end{aligned} \tag{A36}$$

which is $\text{Gamma}(p/2 + c_e, \sum_{l=1}^{p}(e_l - \mu_e)^2/2 + d_e)$.

A.3 Technical details for RJ-MCMC algorithm

Following construction of RJ-MCMC by Green [17], the acceptance probability of the form $\alpha = \min(1, R)$, where

$$R = (\text{Likelihood Ratio}) \cdot (\text{Prior Ratio}) \cdot (\text{Proposal Ratio}) \cdot (\text{Jacobian}). \tag{A37}$$

Since it is drawing new principal components independent of the current parameters, the Jacobian is one, being the determinant of an identity matrix.

A.3.1 Birth move

Under the birth move, the prior ratio is

$$\frac{f(p+1)}{f(p)} = \frac{\frac{e^{-\lambda}\lambda^{p+1}}{(p+1)!}}{\frac{e^{-\lambda}\lambda^{p}}{p!}} = \frac{\lambda}{p+1}. \tag{A38}$$

The proposal ratio is

$$\frac{f(\text{birth}|\text{death}) \cdot f(\text{death})}{f(\text{death}|\text{birth}) \cdot f(\text{birth})} = \frac{\frac{1}{p+1} \cdot f(\text{death})}{1 \cdot f(\text{birth})} = \frac{f(\text{death})}{(p+1)f(\text{birth})}. \tag{A39}$$

Hence the algorithm of birth move is:

(1) Change the number of principal components from p to $p+1$.

(2) The acceptance probability of birth moving is

$$\alpha = \min\left\{1, \frac{L_1}{L_2} \cdot \frac{\lambda}{p+1} \cdot \frac{f(\text{death})}{(p+1)f(\text{birth})}\right\}, \tag{A40}$$

where the likelihood functions L_1 and L_2 are evaluated at the numbers of principal components $p+1$ and p.

(3) Draw a uniform random variable $v \sim U(0,1)$.

(4) If $v < \alpha$, then accept $p+1$; otherwise, keep the number of principal components as p.

A.3.2 Death move

Under the death move, the prior ratio is

$$\frac{f(p-1)}{f(p)} = \frac{\frac{e^{-\lambda}\lambda^{p-1}}{(p-1)!}}{\frac{e^{-\lambda}\lambda^{p}}{p!}} = \frac{p}{\lambda}. \tag{A41}$$

The proposal ratio is

$$\frac{f(\text{death}|\text{birth}) \cdot \mathrm{f}(\text{birth})}{f(\text{birth}|\text{death}) \cdot f(\text{death})} = \frac{1 \cdot f(\text{birth})}{\frac{1}{p} \cdot f(\text{death})} = \frac{pf(\text{birth})}{f(\text{death})}. \tag{A42}$$

Hence the algorithm of death move is:

(1) Change the number of principal components from p to $p-1$.

(2) The acceptance probability of death moving is

$$\alpha = \min\left\{1, \frac{L_1}{L_2} \cdot \frac{p}{\lambda} \cdot \frac{pf(\text{birth})}{f(\text{death})}\right\}, \tag{A43}$$

where the likelihood functions L_1 and L_2 are evaluated at the numbers of principal components p and $p-1$.

(3) Draw a uniform random variable $v \sim U(0,1)$.

(4) If $v < \alpha$, then accept $p-1$; otherwise, keep the number of principal components as p.

Appendix B. Details of the simulation studies

B.1 Simulation setting one

There are 100 simulations conducted for Gaussian longitudinal data. In every simulation study, $\mathcal{N} = 100$ independent individuals are simulated, with $n_i = 15 - 30$ observations irregularly spaced on time interval $\mathcal{T} = [0, 30]$. Each trajectory was sampled at a random number of points 15–30, chosen from a discrete uniform distribution. The locations of the points were randomly chosen from the entire interval $\mathcal{T} = [0, 30]$ without replacement. In the Gaussian setting, the longitudinal process is generated as a Gaussian process $Y_i(t) = X_i(t) + \varepsilon_i(t)$ where $X_i(t)$ is the ith realization of a Gaussian process with a Karhunen-Loève expansion and $X_i(t) = \mu(t) + \psi(t)\xi_i$. Let the mean function $\mu(t)$ be

$$\mu(t) = \sin(2\pi t - 0.6) + \sin(\pi t + 0.4) + 2t, \tag{B1}$$

where different ts are irregularly spaced on time interval $[0, 30]$. The eigenfunctions $\psi = (\psi_1, \ldots, \psi_p)$ are

$$\psi_1(t) = \frac{\cos(\pi t/5)}{\sqrt{5}}, \psi_2(t) = \frac{\sin(\pi t/5)}{\sqrt{5}}, \psi_3(t) = \frac{\sin(2\pi t/5)}{\sqrt{5}}. \tag{B2}$$

The eigenvalues are $d_1 = 5, d_2 = 3, d_3 = 1.5$, and $d_p = 0$ for $p \geq 4$. The principal component scores are simulated as $\xi_i = (\xi_{i1}, \xi_{i2}, \xi_{i3})^T \sim \text{Normal}(0, D_\xi)$ with $D_\xi = \text{diag}(25, 9, 2.25)$. The measurement error $\varepsilon(t)$ is a Gaussian white noise process with variance $\sigma_\varepsilon^2 = 0.7^2$.

B.2 Simulation setting two

There are 100 simulations conducted for binary longitudinal data. In every simulation study of the binary data, $\mathcal{N} = 25$ independent individuals are simulated, with $n_i = 1 - 10$ observations irregularly spaced on time interval $\mathcal{T} = [0, 10]$. Each trajectory was sampled at a random number of points 1–10, chosen from a discrete uniform distribution. The locations of the points were randomly chosen from the entire interval $\mathcal{T} = [0, 10]$ without replacement. In the binary setting, binary responses will be generated by logistic model using the logistic probability distribution function. For the binary trajectories $S_i(t)$, a logit model is $S_i(t) \sim \text{Binomial}\{n_i, P_i(t)\}$ where $\text{Logit}P_i(t) = \mu(t) + \psi(t)\xi_i$. The functions of $\mu(t)$ and $\psi(t)\xi_i$ are the same as those in the Gaussian setting but different ts are irregularly spaced on time interval $[0, 10]$. In the proposed Bayesian method, the parameters are estimated by probit model via MCMC. Defining probit model as $\Phi(\zeta) = P(S_i = 1)$ and logistic model as $\Pi(\zeta) = P(S_i = 1)$, where $\Phi(\cdot)$ is a standard normal cumulative distribution function, $\Pi(\cdot)$ is a logistic function, and ζ is an argument, the approximate relationship between probit model and logistic model is

$$\Phi(\zeta) = \Pi(c \cdot \zeta), \tag{B3}$$

where constant $c = \pi/\sqrt{3}$ [53]. Hence in the comparison of the results from logistic model to probit model, the final estimations of the probit model will multiply the constant value.

B.3 Simulation setting three

There are 150 simulations in total conducted for the Gaussian longitudinal data. The mean function and three eigenfunctions are the same as those in simulation setting one. There are six different simulation studies under $n_i = 1 - 5, 3 - 10, 6 - 15, 9 - 20, 12 - 15$ and 15–30 with fixed $\sigma_\epsilon^2 = 0.7^2$. In each study, $\mathcal{N} = 25$ individuals (data sets) were simulated, with n_i observations irregularly spaced on time interval $\mathcal{T} = [0, 30]$. Each trajectory was sampled at a random number of points 1–5, 3–10, 6–15, 9–20, 12–15 and 15–30, chosen from a discrete uniform distribution. The locations of the points were randomly chosen from the entire interval $\mathcal{T} = [0, 30]$ without replacement.

B.4 Simulation setting four

There are 150 simulations in total conducted for the Gaussian longitudinal data. The mean function and three eigenfunctions are the same as those in simulation setting one. The measurement error $\varepsilon(t)$ is a Gaussian white noise process with variance σ_ϵ^2. There are six different simulation studies under $\sigma_\epsilon^2 = 0.5^2, 0.7^2, 0.9^2, 1.1^2, 1.3^2$ and 1.5^2. In each study, $\mathcal{N} = 25$ individuals (data sets) were simulated, with fixed $n_i = 9 - 20$ observations irregularly spaced on time interval $\mathcal{T} = [0, 30]$. Each trajectory was sampled at a random number of points 9–20, chosen from a discrete uniform distribution. The locations of the points were randomly chosen from the entire interval $\mathcal{T} = [0, 30]$ without replacement.